# Tara 11-12-26 Conversation 1: Lab Report 3 demo slide deck

Claude Code session, 11-11-2026, 18:39 - 11-12-2026 11:48 PST. User messages and Claude replies in full; tool calls and outputs shortened; Claude's internal reasoning omitted.

**User · 18:39**

Build a slide deck with the same template and formatting requirements as the final iteration of the 11-11-26 group meeting presentation. The 1st slide should be a title slide with the title of our project, our names, and "Lab Report 3 Demonstration of Agent System", and an AI usage disclaimer like the 11-11 group meeting one. The 2nd slide should be the full system diagram from 11/4 with all text (it is ok if it is smaller than the required font sizes from the 11/11 presentation). In the slide title, say all modules have been built and tested Then there should be slides showing demonstrations of our components: the catalogs, the OpenSeesPy write/run pipeline, the Abaqus verification loop, and the full end-to-end baseline run. These should roughly follow the screenshot of the slide above.
Use the outline Demo Slides Outline in F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3

**update_plan** task list

```
[in_progress] Find outline, 11-11 deck, 11-4 diagram, demo artifacts
[pending] Read pptx skill and 11-11 deck template
[pending] Gather inputs/outputs for each component demo
[pending] Build the deck
[pending] Render and visually check slides
```

```
Plan shown (5 steps).
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls -la "Lab Report 3"; find . -iname "*.pptx" -not -path "./.git/*" | head
-50; ls "Code for Project"
```

```
total 1076
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 18:38 .
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 18:38 ..
-rw-r--r-- 1 tvenkata 1049089 1094833 Nov 11 18:38 Demo Slides Outline.docx
./11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx
10-22-26 (Federico)
10-23-26 (Tara)
10-26-26 (Federico and Tara)
10-26-26 (Federico)
10-29-26 (Federico)
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls -la "11-11-26 Group Meeting Presentation"; ls -R "Code for Project/11-4-26
(Tara)" | head -60; ls "Code for Project/11-11-26 (Tara)"
```

**Skill**

```
{
 "skill": "anthropic-skills:pptx"
}
```

```
total 2568
drwxr-xr-x 1 tvenkata 1049089       0 Nov  7 10:59 .
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 18:38 ..
-rw-r--r-- 1 tvenkata 1049089 2618558 Nov  7 10:59 AI Agents for Thin-Shell Composite Booms (11-11-26).pptx
drwxr-xr-x 1 tvenkata 1049089       0 Nov  7 10:59 Lab Reports
drwxr-xr-x 1 tvenkata 1049089       0 Nov  7 10:59 class notes
drwxr-xr-x 1 tvenkata 1049089       0 Nov  7 10:59 similar presentations
Code for Project/11-4-26 (Tara):
abaqus_plate_tip_load_laminate.py
abaqus_plate_tip_load_laminate_instructions.txt
[... shortened ...]
```

```
Launching skill: anthropic-skills:pptx
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; $PY -c "import pptx, docx; print('ok')"; which python node soffice pdftoppm 2>&1; S="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx"; $PY "$S/scripts/slide_text.py" "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx"
```

```
Exit code 1
Traceback (most recent call last):
  File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'pptx'
which: no node in (/n//bin:/ucrt64/bin:/usr/local/bin:/usr/bin:/bin:/ucrt64/bin:/usr/bin:/n/bin:/c/Program Files/Python313/Scripts:/c/Program Files/Python313:/c/SIMULIA/Commands2026:/c/Program Files/NVIDIA GPU Computing Toolkit/CUDA/v12.5/bin:/c/Program Files/NVIDIA GPU Computing Toolkit/CUDA/v12.5/libnvvp:/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/c/Program Files (x86)/Common Files/Intel/Shared Libraries/bin32:/c/Program Files (x86)/Common Files/Intel/Shared Libraries/bin:/c/SIMULIA/Commands/2024:/c/Program Files/Microsoft MPI/Bin:/c/SIMULIA/Commands:/c/Windows/system32:/c/Windows:/c/Windows/System32/Wbem:/c/Windows/System32/WindowsPowerShell/v1.0:/c/Windows/System32/OpenSSH:/c/Program Files/dotnet:/c/Program Files/Dell/SysMgt/oma/bin:/c/Program Files/Dell/SysMgt/shared/bin:/c/Program Files/Dell/SysMgt/sm/dellvl:/c/Program Files/Dell/SysMgt/sm:/c/Program Files/Dell/SysMgt/OM_iDRACTools/racadm:/c/Program Files/MATLAB/R2026a/runtime/win64:/c/Program Files/MATLAB/R2026a/bin:/c/Program Files/MATLAB/R2025b/bin:/c/Program Files/MATLAB/R2022a/runt
[... shortened ...]
```

**Bash**

```
python -c "import pptx; print(pptx.__version__)" 2>&1; python -c "import docx" 2>&1; ls "/c/Program Files/Microsoft Office/root/Office16/POWERPNT.EXE" "/c/Program Files/LibreOffice/program/soffice.exe" 2>&1; ls /c/Users/tvenkata/miniforge3/envs/
```

```
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import pptx; print(pptx.__version__)
    ^^^^^^^^^^
ModuleNotFoundError: No module named 'pptx'
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import docx
ModuleNotFoundError: No module named 'docx'
ls: cannot access '/c/Program Files/Microsoft Office/root/Office16/POWERPNT.EXE': No such file or directory
[... shortened ...]
```

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; cd "$SP"; python -m venv venv && ./venv/Scripts/python.exe -m pip install -q python-pptx python-docx pymupdf pillow defusedxml lxml 2>&1 | tail -3; ls "/c/Program Files (x86)/Microsoft Office" 2>&1 | head
```

```
[notice] A new release of pip is available: 26.2.1 -> 26.3
[notice] To update, run: C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-3c5983a9a11f\scratchpad\venv\Scripts\python.exe -m pip install --upgrade pip
ls: cannot access '/c/Program Files (x86)/Microsoft Office': No such file or directory
Shell cwd was reset to F:\Tara\BenazzoVenkatadri_ThinShellSims
```

Claude · 18:40

Setting up tools (python-pptx installed in a scratch venv); now reading the outline and the 11-11 deck.

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; S="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx"; cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; $PY "$S/scripts/slide_text.py" "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" > "$SP/deck1111.md"; wc -c "$SP/deck1111.md"; $PY - <<'EOF'
import docx
d=docx.Document("Lab Report 3/Demo Slides Outline.docx")
for p in d.paragraphs:
    if p.text.strip(): print(f"[{p.style.name}] {p.text}")
for t in d.tables:
    print("TABLE")
```

```
    for r in t.rows: print(" | ".join(c.text for c in r.cells))
print("images:", len(d.inline_shapes))
EOF
```

```
27210 C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad/deck1111.md
[No Spacing] System diagram (take system_diagram_verification.png from 11-4-26 code folder):
[No Spacing] On the slide, put in the title that all modules have been built and tested, including the agents
[No Spacing] For each component: Put screenshot of the component being tested, describe what that component does, screenshot
of prompt given, screenshot of output provided, say it's working and any failure cases
[No Spacing] Each of the catalogs (leave these as placeholders for now):
[No Spacing] Screenshot of the corresponding green box in the system diagram
[No Spacing] Screenshot of Federico's prompt to create the catalog
[No Spacing] Table with a few sample lines from each catalog
[No Spacing] OpenSeesPy write/run pipeline:
[No Spacing] Screenshot of bottom left quadrant of system diagram
[... shortened ...]
```

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
PY="$SP/venv/Scripts/python.exe"; cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; mkdir -p "$SP/docimg"; $PY - <<EOF
import docx
from docx.oxml.ns import qn
from PIL import Image
import io
d=docx.Document("Lab Report 3/Demo Slides Outline.docx")
n=0
for i,p in enumerate(d.paragraphs):
    blips=p._p.findall('.//'+qn('a:blip'))
    for b in blips:
        rid=b.get(qn('r:embed'))
        part=d.part.related_parts[rid]
        n+=1
        ext=part.partname.split('.')[-1]
        fn=f"$SP/docimg/img{n:02d}.{ext}"
        open(fn,'wb').write(part.blob)
        im=Image.open(io.BytesIO(part.blob))
        print(f"img{n:02d} after para {i}: '{p.text[:60]}' size={im.size}")
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/deck1111.md

```
img01 after para 1: '' size=(722, 589)
img02 after para 10: '' size=(722, 589)
img03 after para 13: '' size=(637, 224)
img04 after para 16: '' size=(744, 145)
img05 after para 18: '' size=(685, 170)
img06 after para 21: '' size=(712, 492)
img07 after para 23: '' size=(768, 591)
img08 after para 26: '' size=(722, 589)
img09 after para 29: '' size=(633, 205)
img10 after para 31: '' size=(769, 240)
[... shortened ...]
```

```
1       <!-- Slide number: 1 -->
2       ![title_bg.png](Picture 1)
3       AI Agents for the Optimization of
4       Thin-Shell Composite Booms
5       Federico Benazzo and Tara Venkatadri
6       Group Meeting Presentation
7       November 11th, 2026
8       Disclaimer: These slides were generated by Claude Opus 5.5 (Anthropic) based on an outline provided by the authors.
9
10      ### Notes:
[... shortened ...]
```

**Bash**

```
ls "/c/Program Files" | grep -iE "office|libre|onlyoffice|wps"; ls "/c/Program Files (x86)" | grep -iE "office|libre"; cd
"F:/Tara/BenazzoVenkatadri_ThinShellSims/11-11-26 Group Meeting Presentation"; find . -type f | head -40
```

```
./AI Agents for Thin-Shell Composite Booms (11-11-26).pptx
./class notes/Ae 165a Lab 2 Analytical Calculations (Appendix D).pdf
./class notes/Lecture_10_Slides.pdf
./class notes/Lecture_1_Slides.pdf
./class notes/Lecture_2_Slides.pdf
./class notes/Lecture_3_Slides.pdf
./class notes/Lecture_4_Slides.pdf
./class notes/Lecture_5_Slides.pdf
./class notes/Lecture_6_Slides.pdf
./class notes/Lecture_7_Slides.pdf
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -rl --include=*.py --include=*.js --include=*.md -iE "title_bg|11-11-26\)\.pptx|caltech_logo" . 2>/dev/null | grep -v "^./.git" | head; ls "Running Log"; git log --oneline -- "11-11-26 Group Meeting Presentation" | head
```

```
Log Before Lab 1 Deadline.txt
Log Before Lab 2 Deadline.txt
Log before Lab 3 Deadline.txt
Orchestrator tool changes.txt
ToDo_project.txt
fc04595 Added 11/11 PPT
ac725c4 Rearranged folder for 11-11 presentation
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -n -iE "presentation|slide|font|11-11|11/11" "Running Log/Log before Lab 3 Deadline.txt" | head -40; grep -n -iE "slide|font|presentation" "Running Log/ToDo_project.txt" | head
```

```
49:Federico's and my advisor (Professor Pellegrino) asked us to give a presentation on our project for the class at our weekly group meeting on November 11th.  We are using this as a preliminary version of the final presentation we will give in class on December 2nd.  I used Claude to design the slides based on a detailed outline and slide template, along with several examples of Federico's and my past presentations (i.e. few-shot prompting).  After several iterations, I obtained a satisfactory set of slides, although some of them are still placeholders because Federico is currently running an initial optimization test that we want to include in the slides eventually.
55:11/11/26
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; sed -n 40,120p "Running Log/Log before Lab 3 Deadline.txt"
```

```
- tools/study.py: state machine and runner. It refuses steps out of order, writes the case files, runs the skills, checks the FE tip stiffness against thin-walled beam theory (15% tolerance), corrects the screening estimates with the FE results, schedules mesh-convergence checks and Abaqus verifications (non-blocking: the loop continues while we run Abaqus), stops on diminishing returns or after 100 iterations, and lists the files to commit.
- hooks/protocol_gate.py, registered in .claude/settings.json: while a study is active, it adds the protocol summary and study state to every action and blocks the forbidden ones.
- The rewritten ThinShellOrchestrator agent and a /design-boom skill (entry point: whoever opens the repo types /design-boom with the requirements); a short CLAUDE.md at the repo root routes boom prompts there. The previous orchestrator stays in .claude/agents_v2.0.
Tests: two studies run to completion in a temporary folder. A 300 mm tape spring with a stiffness requirement and a 30 mm hub took 10 s per design; the FE stiffness was 1.7% from beam theory, the finer mesh changed it by 0.5%, and the Abaqus script with instructions was generated. A 200 mm tape spring with
[... shortened ...]
```

**Claude · 18:41**

Inspecting the 11-11 deck's XML (fonts, sizes, colors, layout positions) since no slide renderer is installed here.

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; $PY - <<'EOF'
from pptx import Presentation
from pptx.util import Emu
p=Presentation("11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx")
print("size",p.slide_width/914400,p.slide_height/914400)
print("layouts",[l.name for l in p.slide_layouts])
for idx in [0,1,8,11,12,18]:
    s=p.slides[idx]; print("=== slide",idx+1,"layout",s.slide_layout.name)
```

```
    for sh in s.shapes:
        t=sh.text_frame.text[:40].replace("\n","/") if sh.has_text_frame else ""
        fonts=set()
        if sh.has_text_frame:
            for para in sh.text_frame.paragraphs:
                for r in para.runs:
                    c=None
                    try: c=r.font.color.rgb if r.font.color and r.font.color.type else None
                    except: pass
                    fonts.add((r.font.name,r.font.size.pt if r.font.size else None,r.font.bold,str(c)))
[... shortened ...]
```

```
Exit code 1
size 13.33299978127734 7.5
layouts ['Title Slide', 'Title and Content', 'Section Header', 'Two Content', 'Comparison', 'Title Only', 'Blank', 'Content
with Caption', 'Picture with Caption', 'Title and Vertical Text', 'Vertical Title and Text']
=== slide 1 layout Blank
  PICTURE (13) 'Picture 1' x=0.00 y=0.00 w=13.33 h=7.50 fill=None set() ''
  TEXT_BOX (17) 'TextBox 2' x=1.20 y=2.35 w=10.93 h=1.45 fill=None {('Calibri', 42.0, True, '000000')} 'AI Agents for the
Optimization of/Thin-S'
  TEXT_BOX (17) 'TextBox 3' x=2.40 y=3.95 w=8.53 h=2.45 fill=None {('Calibri', 16.0, False, '595959'), ('Calibri', 24.0,
False, '000000')} 'Federico Benazzo and Tara Venkatadri/Gro'
=== slide 2 layout Blank
  AUTO_SHAPE (1) 'Rectangle 1' x=0.00 y=0.01 w=13.33 h=0.61 fill=F2F2F2 set() ''
  LINE (9) 'Connector 2' x=0.00 y=0.62 w=13.33 h=0.00 fill=None set() ''
[... shortened ...]
```

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; cd
"$SP"; rm -rf d1111; mkdir d1111; cd d1111; unzip -q -o "F:/Tara/BenazzoVenkatadri_ThinShellSims/11-11-26 Group Meeting
Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx"; ls ppt/media; python - <<'EOF'
import re
x=open('ppt/slides/slide12.xml',encoding='utf8').read()
print(len(x)); print(x[:200])
# table props
i=x.find('<a:tbl>'); print(x[i:i+3000])
EOF
```

```
Exit code 1
image1.png
image10.png
image11.png
image12.jpg
image13.png
image2.png
image3.png
image4.png
image5.png
[... shortened ...]
```

**Bash**

```
cd "C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/d1111";
export PYTHONIOENCODING=utf-8; python - <<'EOF'
x=open('ppt/slides/slide12.xml',encoding='utf8').read()
print(x)
EOF
```

```
<?xml version='1.0' encoding='UTF-8' standalone='yes'?>
<p:sld xmlns:a="http://schemas.openxmlformats.org/drawingml/2006/main"
xmlns:p="http://schemas.openxmlformats.org/presentationml/2006/main"
xmlns:r="http://schemas.openxmlformats.org/officeDocument/2006/relationships"><p:cSld><p:spTree><p:nvGrpSpPr><p:cNvPr id="1"
name=""/><p:cNvGrpSpPr/><p:nvPr/></p:nvGrpSpPr><p:grpSpPr/><p:sp><p:nvSpPr><p:cNvPr id="2" name="Rectangle 1"/><p:cNvSpPr/>
<p:nvPr/></p:nvSpPr><p:spPr><a:xfrm><a:off x="0" y="12801"/><a:ext cx="12191695" cy="558698"/></a:xfrm><a:prstGeom
prst="rect"><a:avLst/></a:prstGeom><a:solidFill><a:srgbClr val="F2F2F2"/></a:solidFill><a:ln><a:noFill/></a:ln>
<a:effectLst/></p:spPr><p:style><a:lnRef idx="1"><a:schemeClr val="accent1"/></a:lnRef><a:fillRef idx="3"><a:schemeClr
val="accent1"/></a:fillRef><a:effectRef idx="2"><a:schemeClr val="accent1"/></a:effectRef><a:fontRef idx="minor">
<a:schemeClr val="lt1"/></a:fontRef></p:style><p:txBody><a:bodyPr rtlCol="0" anchor="ctr"/><a:lstStyle/><a:p/></p:txBody>
</p:sp><p:cxnSp><p:nvCxnSpPr><p:cNvPr id="3" name="Connector 2"/><p:cNvCxnSpPr/><p:nvPr/></p:nvCxnSpPr><p:spPr><a:xfrm>
```

```
<a:off x="0" y="571500"/><a:ext cx="12191695" cy="0"/></a:
[... shortened ...]
```

Claude · 18:42

Template style captured (Calibri, 36pt titles on a grey header with orange rule, orange table headers, peach callouts, Caltech logo). Now viewing the outline's 21 screenshots.

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img01.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img03.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img04.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img05.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img07.png

```
[image]
```

**Bash**

```
cd "C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg";
md5sum img01.png img02.png img08.png; "../venv/Scripts/python.exe" -c "from PIL import Image;
print(Image.open(r'F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26
(Tara)/system_diagram_verification.png').size)"
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img09.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img10.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img11.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img12.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img13.png

```
527878daa829e9eb90587f527bda0fe2 *img01.png
527878daa829e9eb90587f527bda0fe2 *img02.png
527878daa829e9eb90587f527bda0fe2 *img08.png
(2890, 2414)
Shell cwd was reset to F:\Tara\BenazzoVenkatadri_ThinShellSims
```

```
[image]
```

```
[image]
```

```
[image]
```

```
[image]
```

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img14.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img15.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img16.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img17.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img18.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img19.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img20.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img21.png

```
[image]
```

**Claude · 18:43**

Mapped all 21 screenshots to slides. Now pulling facts from the 11-11 study (how designs were ruled out, Abaqus run details) and sample catalog rows.

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; ls; ls toy_orch tape_spring_L001_000_tip_load | head -50; wc -l SUMMARY.md
```

```
L001_000.json
L001_909090.json
SUMMARY.md
TapeSpring_L001_000_report.pdf
_conv_pdf
_pylibs
ab
abaqus_scratch
build_report.py
coil_hub_check.py
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; cat SUMMARY.md; find toy_orch -maxdepth 4 | head -30; ../../11-11-26*/x 2>/dev/null; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; "$SP/venv/Scripts/python.exe" -c "
import fitz
d=fitz.open('ts_toy_minmass_REPORT.pdf'); print(d.page_count)
t=''.join(p.get_text() for p in d); open(r'$SP/toyreport.txt','w',encoding='utf8').write(t); print(len(t))"
```

```
# 11-11-26 (Tara): tape spring L001 [0/0/0], R = 20 mm, 180 deg, L = 1000 mm, cantilever

Laminate: L001 (IM7/8552), [0/0/0], t_ply 0.035 mm, h = 0.105 mm (`L001_000.json`; OpenSees ABD vs CLT: A, B exact, D -0.4%).
Section: arc R = 20 mm, b = pi R = 62.83 mm (centre of curvature +z, edges up). Root edge x0 clamped, tip xL rigid end.
Mesh: ASDShellQ4, 2.5 mm (400 x 25 = 10000 elements, 62562 DOFs). Python: shells env, 16 BLAS threads.
Hashin 2D (failure-check skill), S23 = Yc/2 default; catalog mean RTD strengths.
```

```
| Case | Script / case | Result | Max Hashin F | Mode, location |
|---|---|---|---|---|
| Axial 10 N (linear) | `ts_L001_axial*.json/_opensees.py` | tip ux = 0.00957 mm (PL/EA 0.00951, +0.6%) | 4.0e-7 (R = 1576)
| fibre tension, uniform s1 = 1.59 MPa |
[... shortened ...]
```

**Bash**

```
cd "C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
PYTHONIOENCODING=utf-8 python -c "print(open('toyreport.txt',encoding='utf8').read()[:14000])"
```

```
ThinShellOrchestrator - toy-catalog
demonstration
Minimum-mass 1 m tape-spring cantilever from the Tape Spring Baseline
catalogs
Study ts_toy_minmass, 11-11-26 (Tara). Generated from the study files by toy_orch_tools/make_report.py.
1. Answer
The lightest boom that meets every requirement is iteration 1: material L001
(IM7/8552), layup Y001 = [0/0/0] (3 plies x 0.035 mm, h = 0.105 mm), geometry
T001 (tape spring, R = 20 mm, 180 deg, flattened width b = 62.83 mm), L = 1000
mm. Mass 10.40 g.
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls "Code for Project/11-11-26 (Tara)/report_figs" "Code for Project/11-11-26
(Tara)/toy_orch_tools" | head -40; for f in .claude/memory.material_property_catalog/UD_lamina_catalog_readable.txt
.claude/memory.layup_catalog/layup_catalog.txt .claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt; do echo "===
$f"; head -40 "$f"; done
```

```
Code for Project/11-11-26 (Tara)/report_figs:
axial.png
bending.png
coiling.png
moment_rotation.png
section.png

Code for Project/11-11-26 (Tara)/toy_orch_tools:
SANDBOX_CHANGES.md
axial_check.py
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -E "^L00[1-3]|^L013|^L024|^L030"
.claude/memory.material_property_catalog/UD_lamina_catalog_readable.txt | cut -c1-260; grep -E "^Y(05[0-9]|1[0-9][0-9]) "
.claude/memory.layup_catalog/layup_catalog.txt | sed -n '1p;40p;120p'; grep -vE "^#"
.claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt | grep -E "^(ID|T013|R011|M008|C006)" | cut -c1-300
```

```
L001   IM7                            Hexcel HexPly 8552 epoxy (NCAMP)             158.5    8.95     4.67      0.316
0.45*   3.09*     1716      2501      285.7    64.0     91.1      0.035      1.58
https://energy.ornl.gov/CFCrush/materials/uou/CAM
L002   IM7                            Hexcel HexPly 8552 epoxy                     171.42   9.08     5.29      0.32
0.45*   3.13*     1200.1    2326.2    199.8    62.3     92.3      0.035      1.58
https://doi.org/10.1016/j.compscitech.2007.02.005
L003   AS4                            Hexcel HexPly 8552 epoxy (NCAMP)             127.2    9.21     4.79      0.302
0.45*   3.18*     1398      1996      267.9    63.9     91.6      0.035      1.58
https://www.wichita.edu/industry_and_defense/NIAR
L013   AS4                            Hercules 3501-6 epoxy (WWFE-I)               126      11       6.6       0.28
0.4     3.93*     1480      1950      200      48       79        0.035      1.58*       https://doi.org/10.1016/S0266-
3538(98)00078-5 ; h
L024   carbon, unspecified            epoxy (lenticular boom CFRP)                 80.08    6.67     2.93      0.344
0.33*   2.50      662.69    1063.4
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -vE "^#" .claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt |
grep -E "^(ID|T001|T013|R011|M008|C006)" | awk '{for(i=1;i<=NF;i++) if(i<=4||i>=10) printf "%s ", $i; print ""}' | cut -c1-
400
```

```
ID type n_arcs w_mm s_B_mm chord_mm width_mm height_mm symmetric flattens origin source
T001 TS 1 0 - - 28.28 5.86 yes yes (single flange) literature https://doi.org/10.1016/j.actaastro.2016.03.029
```

```
T013 TS 1 0 - - 30.31 9.82 yes yes (single flange) literature https://authors.library.caltech.edu/records/znww4-fsn43
R011 TRAC 1+1 8 21.36 - 27.2 21.6 yes yes literature Benazzo & Pellegrino, AIAA SciTech 2025 (local copy
.claude/memory.material_property_catalog/AIAA_SciTech_2025_paper.pdf)
M008 OM 3+3 4.5 120.95 77 86 77 yes yes literature https://ntrs.nasa.gov/citations/20190033365 ;
https://doi.org/10.1016/j.compstruct.2019.111166 ; https://ntrs.nasa.gov/citations/20205010609
C006 CO 11+11 9 222.01 129.98 147.98 127.94 yes yes derived https://ntrs.nasa.gov/citations/20205003429
```

**Claude · 18:43**

Pulled the study facts (19 of 27 designs ruled out by screening, 8 FE iterations, optimum found at iteration 1). Now cropping diagram regions for each component slide.

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; mkdir
-p "$SP/img"; "$SP/venv/Scripts/python.exe" - <<EOF
from PIL import Image, ImageDraw
im=Image.open(r"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26
(Tara)/system_diagram_verification.png").convert("RGB")
s=im.size[0]/722
def crop(box,name,hl=None):
    b=[int(v*s) for v in box]
    c=im.crop(b).copy()
    if hl:
        d=ImageDraw.Draw(c)
        h=[int(v*s) for v in hl]; h=[h[0]-b[0],h[1]-b[1],h[2]-b[0],h[3]-b[1]]
        d.rectangle(h,outline=(237,125,49),width=14)
    c.save(f"$SP/img/{name}.png"); print(name,c.size)
top=(120,50,585,206)
crop(top,"diag_mat",(124,126,276,204))
crop(top,"diag_lay",(284,126,437,204))
crop(top,"diag_geo",(443,126,582,204))
crop((0,296,362,545),"diag_bl")
crop((350,292,722,582),"diag_br")
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/img/diag_bl.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/img/diag_br.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/img/diag_mat.png

```
diag_mat (1861, 624)
diag_lay (1861, 624)
diag_geo (1861, 624)
diag_bl (1449, 997)
diag_br (1490, 1161)
```

```
[image]
```

```
[image]
```

```
[image]
```

**Bash**

```
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
"$SP/venv/Scripts/python.exe" - <<EOF
from PIL import Image, ImageDraw
im=Image.open(r"F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26
(Tara)/system_diagram_verification.png").convert("RGB")
s=im.size[0]/722
def crop(box,name,hl=None,white=None):
    b=[int(v*s) for v in box]
    c=im.crop(b).copy(); d=ImageDraw.Draw(c)
    if white: d.rectangle(white,fill=(255,255,255))
    if hl:
        h=[int(v*s) for v in hl]; h=[h[0]-b[0],h[1]-b[1],h[2]-b[0],h[3]-b[1]]
        d.rounded_rectangle(h,radius=30,outline=(237,125,49),width=12)
    c.save(f"$SP/img/{name}.png"); print(name,c.size)
top=(118,50,592,198)
```

```
crop(top,"diag_mat",(126,128,276,188))
crop(top,"diag_lay",(286,128,436,188))
crop(top,"diag_geo",(445,128,589,188))
crop((0,299,362,545),"diag_bl")
crop((350,297,722,582),"diag_br",white=(0,780,75,920))
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/img/diag_geo.png

```
diag_mat (1897, 592)
diag_lay (1897, 592)
diag_geo (1897, 592)
diag_bl (1449, 985)
diag_br (1490, 1141)
```

```
[image]
```

Claude · 18:45

Diagram crops look right. Checking a couple of remaining facts (Palmeri & Laurenzi citation, generated-script length) before building.

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; ls Literature* 2>/dev/null | grep -i -E "palmeri|laurenzi"; find . -iname
"*palmeri*" -not -path "./.git/*" | head -3; wc -l "Code for Project/11-11-26
(Tara)/tape_spring_L001_000_tip_load/"*_opensees.py; ls "Code for Project/11-11-26
(Tara)/tape_spring_L001_000_tip_load/results"; grep -rn "11.7" "Code for Project/11-11-26
(Tara)/tape_spring_L001_000_tip_load/results" 2>/dev/null | head -3
```

```
<persisted-output>
Output too large (306KB). Full output saved to: C:\Users\tvenkata\.claude\projects\F--Tara-BenazzoVenkatadri-
ThinShellSims\07583a8d-5e34-4ade-abc5-3c5983a9a11f\tool-results\bnk5cg8w6.txt

Preview (first 2KB):
Palmeri and Laurenzi 2024 Neural Network-Based Surrogate Modeling for Buckling Performance Optimization of Lightweight-
Composite Collapsible Tubular Masts.pdf
./Claude Conversations/Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf
./Literature/Palmeri and Laurenzi 2024 Neural Network-Based Surrogate Modeling for Buckling Performance Optimization of
Lightweight-Composite Collapsible Tubular Masts.pdf
  20675 Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load/ts_L001_000_linear_h2p5_opensees.py
   5639 Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load/ts_L001_000_linear_h5_opensees.py
  26314 total
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; "C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad/venv/Scripts/python.exe" -c "
import pymupdf
d=pymupdf.open('Literature/Palmeri and Laurenzi 2024 Neural Network-Based Surrogate Modeling for Buckling Performance
Optimization of Lightweight-Composite Collapsible Tubular Masts.pdf')
print(d[0].get_text()[:1200].encode('ascii','replace').decode())"
```

```
Citation: Palmeri, F.; Laurenzi, S.
Neural Network-Based Surrogate
Modeling for Buckling Performance
Optimization of Lightweight-
Composite Collapsible Tubular Masts.
Biomimetics 2024, 9, 494.
https://doi.org/10.3390/
biomimetics9080494
Academic Editor: Xiaoming Zhang
Received: 9 July 2024
[... shortened ...]
```

**update_plan** task list

```
{"advance": true}
```

```
Plan updated: step 2 of 5 is in progress.
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; grep -E "^Y[0-9]+ +5 +\[45/-45/0/-45/45\]|^Y[0-9]+ +5 +\
[0/45/-45/45/0\]|^Y016" .claude/memory.layup_catalog/layup_catalog.txt; grep -E "yes +yes$"
.claude/memory.layup_catalog/layup_catalog.txt | grep -v "/90\|0/0/0" | head -3; grep -E " no +yes$"
.claude/memory.layup_catalog/layup_catalog.txt | head -3; ls /c/Windows/Fonts | grep -iE "^calibri|seguisym"
```

```
Y016   4         [0/0/0/0]             0.140   1          0           yes         yes
Y053   5         [0/45/-45/45/0]       0.175   3          -45/0/45    no          no
Y123   5         [45/-45/0/-45/45]     0.175   3          -45/0/45    no          yes
Y107   5         [30/-30/0/-30/30]     0.175   3          -30/0/30    no          yes
Y123   5         [45/-45/0/-45/45]     0.175   3          -45/0/45    no          yes
Y139   5         [60/-60/0/-60/60]     0.175   3          -60/0/60    no          yes
calibri.ttf
calibrib.ttf
calibrii.ttf
calibril.ttf
[... shortened ...]
```

**Write** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py

```
"""Build the Lab Report 3 demonstration deck in the style of the 11-11-26 group meeting deck.

Run with any Python that has python-pptx, python-docx and Pillow:
    python "Lab Report 3/build_demo_deck.py"
Inputs: the 11-11-26 deck (theme, title background, Caltech logo, section drawings), the screenshots
in "Lab Report 3/Demo Slides Outline.docx" and the 11-04 system diagram PNG.
Output: "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx".
"""
import copy
import io
import os

[... shortened ...]
```

```
File created successfully at: F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py (file state is current
in your context — no need to Read it back)
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; "$SP/venv/Scripts/python.exe" "Lab Report 3/build_demo_deck.py" &&
PYTHONIOENCODING=utf-8 "$SP/venv/Scripts/python.exe" - <<'EOF'
from pptx import Presentation
from collections import Counter
p=Presentation("11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx")
c=Counter()
def walk(shapes,i):
    for sh in shapes:
        if sh.has_text_frame:
            for para in sh.text_frame.paragraphs:
                for r in para.runs:
                    if r.font.size: c[(r.font.size.pt)]+=1
                    if r.font.size and r.font.size.pt<18 and i not in (0,): print(i+1, r.font.size.pt, repr(r.text[:40]))
        if sh.has_table:
            for row in sh.table.rows:
                for cell in row.cells:
                    for para in cell.text_frame.paragraphs:
                        for r in para.runs:
                            if r.font.size: c[('tbl',r.font.size.pt)]+=1
[... shortened ...]
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
2 12.0 '2'
2 12.0 '[1] Daraio, C. (2026). ME/CE/AM 295 Agen'
2 12.0 '[2] Yao, S., et al. (2023). ReAct: Syner'
3 12.0 '3'
3 12.0 '[3] Murphey, T. W., et al. (2015). High '
3 12.0 '[4] Fernandez, J. M. (2017). Advanced de'
3 12.0 '[5] Murphey, T. W., et al. (2026). Perfo'
3 12.0 '[6] Liu, T.-W., et al. (2023). Folding b'
```

```
4 12.0 '4'
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3"; sed -i 's/{"size": 14, "color": GREY_TXT}/{"color": GREY_TXT}/;
s/size=16 if "," in txt else 18/size=18/; s/\[\[(t, {})\]\], fill=f, line=l, size=16/[[(t, {})]], fill=f, line=l, size=18/'
build_demo_deck.py; grep -n "size=16\|size\": 14" build_demo_deck.py
```

```
[This command modified 1 file you've previously read: Lab Report 3\build_demo_deck.py. Call Read before editing.]
```

**Write** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/render.py

```
"""Approximate PIL renderer of a python-pptx deck for layout QA (no LibreOffice here).
Draws rectangles / rounded boxes, pictures, text wrapped with real Calibri metrics, and tables with
auto-grown rows. Reports text that overflows its box (or table rows that grow past their set height).
"""
import io
import sys

from PIL import Image, ImageDraw, ImageFont
from pptx import Presentation
from pptx.util import Emu

DPI = 110
[... shortened ...]
```

```
File created successfully at: C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad/render.py (file state is current in your context — no need to Read it back)
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; mkdir -p "$SP/r"; "$SP/venv/Scripts/python.exe" "Lab Report 3/build_demo_deck.py"
&& PYTHONIOENCODING=utf-8 "$SP/venv/Scripts/python.exe" "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent
System.pptx" "$SP/r/s"
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 3: text overflow 0.04 in in 'TextBox 9': 'Input / Output Example'
slide 3: text overflow 0.08 in in 'Rounded Rectangle 10': '✓ Working'
slide 3: table 'Table 14' grows 0.82 in past set height, bottom at 6.86 in
slide 4: text overflow 0.04 in in 'TextBox 9': 'Input / Output Example'
slide 4: text overflow 0.08 in in 'Rounded Rectangle 10': '✓ Working'
slide 5: text overflow 0.04 in in 'TextBox 9': 'Input / Output Example'
slide 5: text overflow 0.08 in in 'Rounded Rectangle 10': '✓ Working'
slide 5: table 'Table 14' grows 0.22 in past set height, bottom at 6.26 in
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-02.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-05.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-07.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-09.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

```
[image]
```

Claude · 18:51

Renders look close; fixing table wrapping on slides 3 and 6, tightening a few boxes. Checking the last four slides next.

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-11.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-12.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-13.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-14.png

```
[image]
```

**Read** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py

```
225                 cell.vertical_anchor = MSO_ANCHOR.MIDDLE
226                 is_head = header and i == 0
227                 fill = ORANGE if is_head else WHITE
228                 if fills and not is_head and fills.get((i, j)) is not None:
229                     fill = fills[(i, j)]
230                 elif fills and not is_head and fills.get((i, None)) is not None:
231                     fill = fills[(i, None)]
232                 cell.fill.solid()
233                 cell.fill.fore_color.rgb = fill
234                 tf = cell.text_frame
[... shortened ...]
```

Claude · 18:51

Now applying the layout fixes.

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3";
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
"$SP/venv/Scripts/python.exe" - <<'EOF'
import re
f="build_demo_deck.py"; s=open(f,encoding="utf8").read()
def rep(a,b,count=1):
    global s
    assert s.count(a)>=1, a
    s=s.replace(a,b) if count==0 else s.replace(a,b,count)
# per-row heights
rep("""    gf = slide.shapes.add_table(nr, nc, Inches(x), Inches(y), Inches(sum(col_w)), Inches(row_h * nr))""",
"""    rh = row_h if isinstance(row_h, (list, tuple)) else [row_h] * nr
    gf = slide.shapes.add_table(nr, nc, Inches(x), Inches(y), Inches(sum(col_w)), Inches(sum(rh)))""")
rep("""        tbl.rows[i].height = Inches(row_h)""","""        tbl.rows[i].height = Inches(rh[i])""")
# pills / headings taller
rep("def status_pill(slide, x, y, ok=True, w=1.75, h=0.4, text=None):","def status_pill(slide, x, y, ok=True, w=1.75,
h=0.45, text=None):")
rep("""    return add_text(slide, x, y, w, 0.4, [[(text, {"bold": True})]], size=size, space_after=0)""",
    """    return add_text(slide, x, y, w, 0.45, [[(text, {"bold": True})]], size=size, space_after=0)""")
# slide 2 typo
rep('("Failure case", "omega boom", "14")','("Failure case", "omega boom", "14", )')
[... shortened ...]
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 3: table 'Table 14' grows 0.82 in past set height, bottom at 6.62 in
slide 5: table 'Table 14' grows 0.22 in past set height, bottom at 6.02 in
slide 6: text overflow 0.07 in in 'TextBox 8': 'Writes the material/layup, mesh and load/BC code, '
slide 6: text overflow 0.25 in in 'TextBox 16': 'FE within 1.3% of beam theory; mesh refinement cha'
slide 9: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
slide 14: table 'Table 10' grows 0.76 in past set height, bottom at 5.06 in
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3"; sed -i 's/add_text(s, X, 6.15, 6.4, 0.45, "FE within/add_text(s, X, 6.15, 6.45, 0.6, "FE within/; s/"FE within 1.3% of beam theory; mesh refinement changes it 0.9%"/"Within 1.3% of beam theory; mesh-converged to 0.9%"/' build_demo_deck.py; grep -n "Within 1.3" build_demo_deck.py
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
470:    add_text(s, X, 6.15, 6.45, 0.6, "Within 1.3% of beam theory; mesh-converged to 0.9%",
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

```
[image]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3";
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
"$SP/venv/Scripts/python.exe" - <<'EOF'
f="build_demo_deck.py"; s=open(f,encoding="utf8").read()
old=s[s.index('        [0.75, 2.25, 1.95, 1.95, 0.98],'):s.index('"Suggested presenter: Federico (built the catalogs).')]
new='''        [0.75, 2.35, 1.9, 1.9, 0.98],
        [["ID", "Fiber/matrix", "E1/E2/G12 (GPa)", "Xt/Yt/S12 (MPa)", "\\u03c1 (g/cm\\u00b3)"],
         ["L001", "IM7/8552 epoxy [1]", "158.5/8.95/4.67", "2501/64.0/91.1", "1.58"],
         ["L013", "AS4/3501-6 epoxy [2]", "126/11/6.6", "1950/48/79", "1.58*"],
         ["L024", "Carbon/epoxy [3]", "80.1/6.67/2.93", "1063/66.0/126.8", "1.60"],
         ["L030", "T1100GC/PAEK [4]", "173/9.0/4.03", "3613/90/69.0", "1.59"]],
        '''
s=s.replace(old,new); open(f,"w",encoding="utf8").write(s)
EOF
cd ..; "$SP/venv/Scripts/python.exe" "Lab Report 3/build_demo_deck.py" && PYTHONIOENCODING=utf-8
"$SP/venv/Scripts/python.exe" "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s"
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 3: table 'Table 14' grows 0.42 in past set height, bottom at 6.22 in
slide 5: table 'Table 14' grows 0.22 in past set height, bottom at 6.02 in
slide 6: text overflow 0.07 in in 'TextBox 8': 'Writes the material/layup, mesh and load/BC code, '
slide 9: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
slide 14: table 'Table 10' grows 0.76 in past set height, bottom at 5.06 in
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-11.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-07.png

```
[image]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; S="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx"; sed -i 's/add_table(s, X, 4.75, \[2.15, 2.6, 1.45, 1.65\],/add_table(s, X, 4.75, [2.1, 2.5, 1.4, 1.85],/' "Lab Report 3/build_demo_deck.py";
"$SP/venv/Scripts/python.exe" "Lab Report 3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 "$SP/venv/Scripts/python.exe"
"$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep -v "grows\|slide 1:";
PYTHONIOENCODING=utf-8 "$SP/venv/Scripts/python.exe" "$S/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -15; PYTHONIOENCODING=utf-8 "$SP/venv/Scripts/python.exe" "$S/scripts/slide_text.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" | grep -iE "lorem|TODO|\[insert" ; ls -la "Lab Report 3"
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 6: text overflow 0.07 in in 'TextBox 8': 'Writes the material/layup, mesh and load/BC code, '
All validations PASSED!
total 3036
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 18:52 .
drwxr-xr-x 1 tvenkata 1049089       0 Nov 11 18:38 ..
-rw-r--r-- 1 tvenkata 1049089 1094833 Nov 11 18:38 Demo Slides Outline.docx
-rw-r--r-- 1 tvenkata 1049089 1961958 Nov 11 18:52 Lab Report 3 Demonstration of Agent System.pptx
-rw-r--r-- 1 tvenkata 1049089   44863 Nov 11 18:52 build_demo_deck.py
[This command modified 1 file you've previously read: Lab Report 3\build_demo_deck.py. Call Read before editing.]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-01.png

```
[image]
```

Claude · 18:53

Deck is done and passes validation. Wrapping up.

**wrap_up** wrap-up

```
The 14-slide Lab Report 3 deck is built in the 11-11 style, with a script to rebuild it, and it passes the file validator.
The catalog prompt screenshots are still placeholders. Not yet opened in real PowerPoint: layout was only checked with an
approximate preview.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 18:53

I've built the deck: Lab Report 3 Demonstration of Agent System.pptx. It has 14 slides in the 11-11 style (grey header with orange rule, Calibri, 36 pt titles, Caltech logo, orange-header tables, peach callouts). Text is at least 18 pt except references and slide numbers at 12 pt, which matches the 11-11 deck. The script that builds it is build_demo_deck.py; rerun it after changing the outline or the screenshots.

**Not yet checked in PowerPoint.** This machine has no PowerPoint or LibreOffice, so I checked layout with a rough preview I wrote myself. It was good enough to find and fix overflow and table wrapping, but please flip through the deck in PowerPoint before presenting. It passes the file validator.

**Slides** 1. **Title:** project title, your names, "Lab Report 3 Demonstration of Agent System", and the same AI disclaimer wording as 11-11. 2. **"All Modules Built and Tested, Including the Agents":** the full 11-4 diagram, with a list of which slides demo each part. 3. **Catalogs, slides 3–5 (materials, layups, cross-sections).** Each slide has the green box cropped from the diagram and highlighted, a "What this section does" text, a placeholder box for Federico's prompt, and real sample rows from the catalogs in `.claude/memory.*`. The cross-section slide also has the section drawings. 4. **OpenSeesPy pipeline, slides 6–7:** - Bottom-left quadrant of the diagram, the prompt screenshot, and a note that the generated script (~20,700 lines) is too long to show. - Tip deflection and rotation at both mesh sizes against beam theory, the two plots, and the mass and Hashin index results. 5. **Abaqus loop, slides 8–9:** - Bottom-right quadrant, the prompt, and a callout about running Abaqus safely on the shared server. - The results table (agreement within 0.7%), the side-by-side plots, and why that error is acceptable. 6. **End-to-end baseline, slides 10–13:** - The sandboxed catalogs, grouped by what each option should do: should be optimal, heavier, or should fail. - The tools-only run, alongside the optimization setup. - The orchestrator run compared with the tools-only baseline and with Abaqus. - A grid of all 27 designs showing the 19 ruled out before any FE run, plus the conclusion and next step. 7. **Failure case, slide 14:** the Palmeri and Laurenzi prompt, a red "Failing" status, and a table of what was expected, what came back, the likely cause and the outcome, with the citation.

**Choices I made beyond the outline** - The component slides follow your template loosely. The template's input/output table would be too small to read with screenshots in its cells, so each slide has a diagram crop and description on the left, the prompt screenshot and output on the right, and a Working/Failing status label. - I retyped the result tables from your screenshots as real PowerPoint tables so they stay readable at 18 pt. Prompts are still shown as screenshots. - Slide 13 is extra: you asked to discuss how Claude ruled out designs, and the outline had that plus the conclusion and next step on one slide, which didn't fit. Its numbers come from `ts_toy_minmass_REPORT.pdf`. - Each slide has speaker notes with a suggested presenter and supporting details.

**Still to do** - Paste Federico's catalog prompt screenshots into the dashed boxes on slides 3–5. - On slide 8, check the CPU wording. Your prompt allowed 16 cores, but the project rules cap Abaqus at 8, so I wrote that the wrapper "caps the CPUs" without a number.

User · 19:49

> Slide 3: make sure each reference fits on 1 line so the references don't bleed off the bottom of the slide Slide 3: Mention permissions split for librarian to be a separate agent from orchestrator (only agent with web access) Slide 4: Replace ply thickness h with t Slide 5: Make sure there's no overlap in the images of different cross-sections Slide 6: Say that objective/constraints calculator does Hashin failure on specified loading cases and also flattening for coiling to match deployable structures application of this project Slide 7: Say the 1e-4 Hashin index is for bending to distinguish it from the coiling one Slide 8: Mention the limit on number of CPUs used on shared lab folder Slide 10: Mention that L001 is on slide 3. Mention that the "should fail" ones should trigger material failure, and that heavier is against the objective of having the lightest boom Slide 11: Explain what strength ratio means Slide 13: More explanation as to why some designs were rejected for coiling or stiffness. In next step, add that optimization orchestrator can pre-screen for viable designs and conduct the necessary grid search much faster than the user Slide 14: Delete 11/10 (the date) from "Input". Replace "Input-Output Example" with "Output". Get rid of the row that says "Failure case". Specify that it's continuous variation in geometry rather than discrete variation. Throughout slides, do subscripts for things like sigma_x, u_z, etc, and remove personal pronouns

**update_plan** task list

```
[in_progress] Add subscript support to text helpers
[pending] Apply slide-by-slide edits (3-14)
[pending] Remove personal pronouns throughout
[pending] Rebuild, render and check every slide
```

```
Plan shown (4 steps).
```

**Read** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py

```
100             "br": crop((350, 297, 722, 582), white=(0, 780, 75, 920)),
101         }
102
103     # ---------------------------------------------------------------------------- helpers
104
105     def _runs(p, runs, size, bold, color, italic=False):
106         if isinstance(runs, str):
107             runs = [(runs, {})]
108         for text, o in runs:
109             r = p.add_run()
[... shortened ...]
```

**Read** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py

```
290         media = base_media(prs)
291         shots = outline_images()
292         diag = diagram_crops()
293         logo = media["caltech_logo.png"]
294
295         # drop every slide of the 11-11 deck, keep its masters and theme
296         sldIdLst = prs.slides._sldIdLst
297         for sldId in list(sldIdLst):
298             prs.part.drop_rel(sldId.rId)
299             sldIdLst.remove(sldId)
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3";
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
"$SP/venv/Scripts/python.exe" - <<'EOF'
from PIL import Image
import io
from pptx import Presentation
p=Presentation(r"F:/Tara/BenazzoVenkatadri_ThinShellSims/11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell
Composite Booms (11-11-26).pptx")
for s in p.slides:
    for sh in s.shapes:
        if sh.shape_type==13:
            d=sh._element.nvPicPr.cNvPr.get("descr")
```

```
            if d.startswith("sec_"): print(d, Image.open(io.BytesIO(sh.image.blob)).size)
EOF
```

```
sec_T013.png (825, 373)
sec_R011.png (670, 627)
sec_M008.png (633, 627)
sec_C006.png (659, 627)
sec_C006.png (659, 627)
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3";
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
"$SP/venv/Scripts/python.exe" - <<'EOF'
f="build_demo_deck.py"; s=open(f,encoding="utf8").read()
def rep(a,b):
    global s
    assert s.count(a)==1, ("COUNT",s.count(a),a[:80])
    s=s.replace(a,b)

# --- subscript markup: "_{...}" -> subscript run
rep('''def _runs(p, runs, size, bold, color, italic=False):
    if isinstance(runs, str):
        runs = [(runs, {})]
    for text, o in runs:
        r = p.add_run()
        r.text = text
        f = r.font''','''SUB = re.compile(r"_\\{([^}]*)\\}")


def _runs(p, runs, size, bold, color, italic=False):
[... shortened ...]
```

```
Exit code 2
/usr/bin/bash: -c: line 117: unexpected EOF while looking for matching `''
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3"; git diff --stat -- build_demo_deck.py 2>/dev/null; grep -c "SUB =
re" build_demo_deck.py
```

```
0
```

Claude · 19:51

Shell quoting broke the edit script; nothing changed yet. Writing the edits as a script file instead.

**Write** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/edits2.py

```
import sys
f = sys.argv[1]
s = open(f, encoding="utf8").read()


def rep(a, b):
    global s
    assert s.count(a) == 1, ("COUNT", s.count(a), a[:90])
    s = s.replace(a, b)


# --- subscript markup: "_{...}" -> subscript run
[... shortened ...]
```

```
File created successfully at: C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad/edits2.py (file state is current in your context — no need to Read it back)
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; $PY "$SP/edits2.py" "Lab Report
3/build_demo_deck.py" && sed -i 's/        sz = r.font.size.pt if r.font.size else 18/        sz = r.font.size.pt if
r.font.size else 18\n        if r.font._element.get("baseline"):\n            sz *= 0.65/' "$SP/render.py" && $PY "Lab
```

```
Report 3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s"; PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts/slide_text.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" | grep -inwE "we|our|us|I|my|me|you|your|ours"
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 3: table 'Table 14' grows 0.42 in past set height, bottom at 6.22 in
slide 5: table 'Table 14' grows 0.22 in past set height, bottom at 6.02 in
slide 9: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
slide 11: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
slide 14: table 'Table 10' grows 1.06 in past set height, bottom at 4.76 in
365:Fails stiffness: beam theory gives > 2 cm deflection ([90/90/90]; R = 2 mm since I ∝ R³)
371:Before the first iteration, the orchestrator's design screen checked all 27 combinations with exact or near-exact formulas: coiling with exact CLT + Hashin 2D (the flattened, hub-bent flange has a known uniform curvature state, so no FE is needed) and stiffness with thin-walled beam theory (k = 3 E_x I / L^3, rejected below 0.85 x 0.05 N/mm). 17 fail coiling, 12 fail stiffness (some both): every L003 row (strengths x0.001), every T002 row (R = 2 mm) and [0]10 on R = 20 mm fai
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; sed -i 's/from the ones our skills set up/from the ones the existing skills set up/' "Lab Report 3/build_demo_deck.py"; grep -nwiE "our|we|us|you|your" "Lab Report 3/build_demo_deck.py" | grep -v "^\s*#" | head; $PY "Lab Report 3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" >/dev/null
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
[This command modified 1 file you've previously read: Lab Report 3\build_demo_deck.py. Call Read before editing.]
```

**update_plan** task list

```
{"done": [1, 2, 3], "start": 4}
```

```
Plan updated: step 4 of 4 is in progress.
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-05.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-13.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-14.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-11.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
[image]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; F="Lab Report 3/build_demo_deck.py"; sed -i 's/"I \\u221d R\\u00b3)"/"I scales as R\\u00b3)"/; s/I ∝ R³)/I scales as R³)/; s/"for coiling (the deployable-boom application), for the orchestrator. Draws the results for the "/"for coiling (the deployable-boom application), for the orchestrator. Draws the results."/; s/^                "user.")$/                )/' "$F"; grep -n "scales as\|Draws the results" "$F"; $PY "$F" && PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep -v grows; PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
```

```
5ac22acf505e/skills/pptx/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --
original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" | tail -2
```

```
481:                 "for coiling (the deployable-boom application), for the orchestrator. Draws the results."
760:                                       "I scales as R³)", {})]], 0.66)]
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 11: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
All validations PASSED!
[This command modified 1 file you've previously read: Lab Report 3\build_demo_deck.py. Call Read before editing.]
```

**Read** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/build_demo_deck.py

```
474                      "its outputs are shown.\n"
475                      "Beam theory: PL^3/3EI with EI = 3.96e7 N mm^2 gives 8.414 mm; FE (2.5 mm mesh) is 1.3% softer; "
476                      "halving the element size changes the tip deflection by 0.9%. Tip rotation within 0.3% of
PL^2/2EI.")
477         _, _, h = add_pic(s, diag["bl"], 0.3, 0.85, w=4.2, name="diagram_section")
478         what_it_does(s, 0.3, 0.85 + h + 0.12, 4.6, 3.0,
479                      "Writes the material/layup, mesh and load/BC code, joins it into one OpenSeesPy script and runs "
480                      "it. Computes the mass and the Hashin failure index for the specified load cases, plus flattening "
481                      "for coiling (the deployable-boom application), for the orchestrator. Draws the results."
482                      )
483         X, W = 5.15, 7.88
```

**Claude** · 19:53

Code is fine; checking slides 6 and 13 one last time.

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-07.png

```
[image]
```

**wrap_up** wrap-up

```
Every requested change is in the deck and the build script, and the deck passes the file validator. Not yet opened in real
PowerPoint: the rough preview here can't show subscripts correctly, so check those when you flip through.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

**Claude** · 19:53

I made all the changes in the deck and in build_demo_deck.py. It passes the file validator, but I haven't opened it in real PowerPoint.

**Check the subscripts in PowerPoint.** The rough preview I use here draws them raised, like superscripts, so I couldn't confirm how they look. In the file they are set as proper subscripts.

- **Throughout:**
- **Subscripts:** σ_xx, u_z, σ_1, τ_12, E_1/E_2/G_12, X_t/Y_t/S_12 and [0]_10 are now real subscripts. In the script you can write one as `σ_{xx}`.
- **Pronouns:** removed from the slides and the speaker notes; for example "our lab server" is now "the lab server". The "I" left on slide 13 is the moment of inertia.
- **Slide 3:**
- **References:** the four references now take one line each, sit above the logo, and stay on the slide. I wrote out the full titles for [2] and [3], since they now have room.
- **Description:** now says the librarian is separate from the orchestrator because of a permissions split: it is the only agent with web access.
- **Slide 4:** the thickness column is now "t (mm)", and the description says "thickness t".
- **Slide 5:** the four cross-section drawings are placed side by side with a fixed gap, scaled to fit the column, so they no longer overlap.

- **Slide 6:** the description says the calculator computes mass and the Hashin failure index for the specified load cases, plus flattening for coiling (the deployable-boom application).
- **Slide 7:** the $4.2\times10^{-4}$ tile now reads "Max Hashin index in bending, 1 N tip load".
- **Slide 8:** the callout now gives the CPU limit: at most 16 cores in total, Abaqus at most 8 through a wrapper that enforces it, and nothing written outside the project repository folder.
- **Slide 10:** the table says "L001 (slide 3)". The bullets explain:
- "Heavier" works against the objective of the lightest boom.
- "Should fail" options should trigger material failure. I noted that [90/90/90] also misses the deflection limit, because the study rejected it for stiffness rather than material failure.
- **Slide 11:** added "Strength ratio R = 1/√F: factor by which the load can grow before Hashin failure (F = 1)". This matches the table (for example, F = 0.183 gives R = 2.34).
- **Slide 13:**
- **Legend:** now explains each rejection. C means flattening plus the 4 cm hub gives Hashin F ≥ 1 (too thick, R too tight, or weak material). S means beam theory gives more than 2 cm deflection ([90/90/90], or R = 2 mm because I scales as $R^3$).
- **Next step:** now adds that the orchestrator pre-screens for viable designs and runs the grid search much faster than a human.
- **Slide 14:** the input label is just "Input: prompt", the heading is "Output", and the "Failure case" row is gone. The likely cause now says "continuous variation of the geometry instead of discrete catalog rows".

User · 19:53

Update the system diagram on slide 2 so the 2 purple blocks where the user moves things from one machine to another are eliminated and ODB extraction is no longer necessary, since Claude is now on the server with Abaqus

**update_plan** task list

```
[in_progress] Read the 11-4 diagram generator
[pending] Make updated diagram without purple move/extract steps
[pending] Use new diagram on slides 2 and 8, rebuild
[pending] Render and check
```

```
Plan shown (4 steps).
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26 (Tara)"; wc -l system_diagram.py; cat system_diagram.py
```

```
137 system_diagram.py
"""
Revised system diagram (after Lab 2, Fig. 1) with the new verification protocol:
the Abaqus verifier agent is split into a deterministic OpenSeesPy -> Abaqus
translator (opensees-to-abaqus skill, with its own model check) and an
independent results-comparison agent that sees only the outputs of the two runs.

  python system_diagram.py [out.png]
"""
import sys
[... shortened ...]
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; cp "Code for Project/11-4-26 (Tara)/system_diagram.py" "Lab Report
3/system_diagram_lab3.py"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad"; cat > "$SP/e3.py" <<'PYEOF'
import sys
f = sys.argv[1]; s = open(f, encoding="utf8").read()
def rep(a, b):
    global s
    assert s.count(a) == 1, a[:60]; s = s.replace(a, b)
```

```
rep('''"""
Revised system diagram (after Lab 2, Fig. 1) with the new verification protocol:''', '''"""
Lab Report 3 version of the 11-04 system diagram (Code for Project/11-4-26 (Tara)/system_diagram.py).
Claude now runs on the lab server that has Abaqus, so the two human steps that moved the script to an
Abaqus machine and the ODB extract back are gone, and no ODB extraction to plain arrays is needed:
the comparison agent reads the Abaqus ODB directly.

Original description: revised system diagram (after Lab 2, Fig. 1) with the new verification protocol:''')
rep('''box(91, 21, 117, 32, 'human', text='Script moved to an\\nAbaqus machine;\\nuser checks set-up', fs=FS - 0.5)
arrow([(104, 40), (104, 32)])
box(109, 4, 144, 15, 'exec', text='Abaqus run, ODB output\\nand ODB extraction to\\nplain arrays', fs=FS - 0.5)
arrow([(104, 21), (104, 9.5), (109, 9.5)])
[... shortened ...]
```

```
Exit code 1
Traceback (most recent call last):
  File "C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\e3.py", line 14, in <module>
    rep('''box(91, 21, 117, 32, 'human', text='Script moved to an\nAbaqus machine;\nuser checks set-up', fs=FS - 0.5)
    ~~~^^^
```

```
import sys
f = sys.argv[1]
s = open(f, encoding="utf8").read()


def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)


rep('''DIAGRAM = os.path.join(ROOT, "Code for Project", "11-4-26 (Tara)", "system_diagram_verification.png")''',
[... shortened ...]
```

```
File created successfully at: C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e4.py (file state is current in your context — no need to Read it back)
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; $PY "$SP/e4.py" "Lab Report 3/build_demo_deck.py" && $PY "Lab Report 3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep -v grows; PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" | tail -1
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 11: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
All validations PASSED!
```

**update_plan** task list

```
{"advance": true}
```

```
Plan updated: step 4 of 4 is in progress.
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-02.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
[image]
```

**Claude · 19:55**

The trimmed diagram is wider, so on slide 2 it now runs under the right-hand panel. Fitting it by width instead.

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; F="Lab Report 3/build_demo_deck.py"; grep -n 'add_pic(s, diag\["full"\]' "$F"; sed -i 's/    add_pic(s, diag\["full"\], 0.3, 0.78, h=6.0, border=False, name="system_diagram")/    add_pic(s, diag["full"], 0.3, 0.95, w=7.3, border=False, name="system_diagram")/' "$F"; grep -n 'add_pic(s, diag\["full"\]' "$F"; $PY "$F" && PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep -v "grows\|slide 1:\|slide 11:"
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-02.png

```
352:    add_pic(s, diag["full"], 0.3, 0.78, h=6.0, border=False, name="system_diagram")
352:    add_pic(s, diag["full"], 0.3, 0.95, w=7.3, border=False, name="system_diagram")
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
[This command modified 1 file you've previously read: Lab Report 3\build_demo_deck.py. Call Read before editing.]
```

```
[image]
```

**wrap_up** wrap-up

```
The system diagram is updated: the two purple file-moving steps and the ODB extraction step are gone. Abaqus now runs on the lab server and the comparison agent reads the ODB directly. The new diagram is on slides 2 and 8. The deck passes the file validator but hasn't been opened in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 19:56

I updated the system diagram and put the new version on slide 2. Slide 8 uses a crop of the same diagram, so I updated that crop too; otherwise it would still show the removed steps.

**What changed in the diagram:** - **Removed:** the two purple steps where you moved the script to an Abaqus machine and the ODB extract back, and the ODB-to-plain-arrays extraction. - **New path:** the translator now feeds one orange box, "Abaqus run on the lab server, by Claude (at most 8 CPUs)". An arrow labeled "ODB" goes from it straight to the results comparison agent. - **Unchanged:** everything else, including the "User reads comparison, adjusts orchestrator" step, which is still a human action.

**Files:** - **Diagram script:** system_diagram_lab3.py writes system_diagram_lab3.png. It's a copy of the 11-4 script with only the verification part changed. The original 11-4 script and image are untouched. - **Deck:** the deck now uses the new image. I trimmed the empty space at the bottom and sized the diagram on slide 2 to fit beside the "Demonstrated on the next slides" panel. - **Speaker notes:** slides 2 and 8 now describe the change.

The deck passes the file validator; I checked the layout with my rough preview, not in PowerPoint.

User · 19:59

Make the yellow box symmetrically between the translator and verification agent

Bash

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3"; grep -n "Abaqus run on the\|ODB'\|109.5, 40"
system_diagram_lab3.py
```

```
124:box(94, 19, 125, 30, 'exec', text='Abaqus run on the\nlab server, by Claude\n(at most 8 CPUs)', fs=FS - 0.5)
125:arrow([(109.5, 40), (109.5, 30)])
130:arrow([(125, 24.5), (155, 24.5), (155, 41)], 'ODB', (140, 24.5), ha='center')
```

Bash

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3"; sed -i "124s/box(94, 19, 125, 30,/box(114, 19, 143, 30,/;
125s/.*/arrow([(109.5, 40), (109.5, 24.5), (114, 24.5)])/; 130s/.*/arrow([(143, 24.5), (147.5, 24.5), (147.5, 41)], 'ODB',
(147.5, 33), ha='center')/" system_diagram_lab3.py; sed -n 120,131p system_diagram_lab3.py;
/c/Users/tvenkata/miniforge3/envs/shells/python.exe system_diagram_lab3.py
```

Read F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/system_diagram_lab3.png

```
# verification protocol
box(94, 40, 125, 55, 'tool', text='OpenSeesPy-to-Abaqus\ntranslator to produce\nAbaqus Python script')
arrow([(77, 47.5), (94, 47.5)], 'assembled\nscript', (88.5, 47.5), ha='center')
arrow([(109.5, 70), (109.5, 55)], 'verify', (109.5, 62.5), ha='center')
box(114, 19, 143, 30, 'exec', text='Abaqus run on the\nlab server, by Claude\n(at most 8 CPUs)', fs=FS - 0.5)
arrow([(109.5, 40), (109.5, 24.5), (114, 24.5)])

box(127.5, 41, 167.5, 56.5, 'agent', 'Results comparison agent',
    'sees only the outputs of both runs;\ndraws both and compares displacements,\nstrains, curvatures, ply stresses,\n'
    'section forces and load path', fs=FS - 1.5)
[... shortened ...]
```

```
[image]
```

Claude · 20:00

Yellow box is now centred between the translator and the comparison agent, with matching L-shaped arrows on each side. Rebuilding the deck.

Bash

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; $PY "Lab Report 3/build_demo_deck.py" &&
PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep -
v "grows\|slide 1:\|slide 11:"; PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-
```

```
sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
5ac22acf505e/skills/pptx/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --
original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" | tail -1
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
All validations PASSED!
```

```
[image]
```

**wrap_up** wrap-up

```
The yellow Abaqus box is now centred between the translator and the results comparison agent, with mirror-image arrows on
each side. The deck was rebuilt with the new diagram on slides 2 and 8 and passes the file validator. It hasn't been opened
in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 20:00

The yellow "Abaqus run on the lab server" box is now centred between the translator and the results comparison agent. The arrow into it comes from the translator in an L-shape, and the "ODB" arrow out of it goes up to the comparison agent in the mirror-image L.

I changed system_diagram_lab3.py, regenerated the image, and rebuilt the deck. Both slide 2 (full diagram) and slide 8 (its crop) show the new layout. The deck passes the file validator, but I haven't opened it in PowerPoint.

User · 20:01

Get rid of ", by Claude"

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe"; sed -i "s/text='Abaqus run on the\\\\nlab server,
by Claude\\\\n(at most 8 CPUs)'/text='Abaqus run on the\\\\nlab server\\\\n(at most 8 CPUs)'/" "Lab Report
3/system_diagram_lab3.py"; grep -n "Abaqus run on the" "Lab Report 3/system_diagram_lab3.py";
/c/Users/tvenkata/miniforge3/envs/shells/python.exe "Lab Report 3/system_diagram_lab3.py" && $PY "Lab Report
3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-
sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
5ac22acf505e/skills/pptx/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --
original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" | tail -1;
PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s"
>/dev/null
```

```
124:box(114, 19, 143, 30, 'exec', text='Abaqus run on the\nlab server, by Claude\n(at most 8 CPUs)', fs=FS - 0.5)
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\system_diagram_lab3.png
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
All validations PASSED!
```

Claude · 20:01

Sed didn't match; using an exact edit instead.

**Read** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/system_diagram_lab3.py

```
124     box(114, 19, 143, 30, 'exec', text='Abaqus run on the\nlab server, by Claude\n(at most 8 CPUs)', fs=FS - 0.5)
```

**Edit** F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/system_diagram_lab3.py

```
- lab server, by Claude\n
+ lab server\n
```

**Bash**

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-
5e34-4ade-abc5-3c5983a9a11f/scratchpad"; PY="$SP/venv/Scripts/python.exe";
/c/Users/tvenkata/miniforge3/envs/shells/python.exe "Lab Report 3/system_diagram_lab3.py" && $PY "Lab Report
3/build_demo_deck.py" && PYTHONIOENCODING=utf-8 $PY "C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-
sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
```

```
5ac22acf505e/skills/pptx/scripts/office/validate.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" --
original "11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" | tail -1;
PYTHONIOENCODING=utf-8 $PY "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s"
>/dev/null
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
The file F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3/system_diagram_lab3.py has been updated successfully.
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\system_diagram_lab3.png
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 14 slides
All validations PASSED!
```

```
[image]
```

**wrap_up** wrap-up

```
The yellow box now reads "Abaqus run on the lab server (at most 8 CPUs)", without ", by Claude". The diagram and deck are
regenerated, slides 2 and 8 show the change, and the file validator passes. Not yet opened in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 20:02

I removed ", by Claude". The yellow box now reads "Abaqus run on the lab server (at most 8 CPUs)".

I regenerated the diagram and rebuilt the deck, so the change shows on both slide 2 and the slide 8 crop. The deck passes the file validator, but I haven't opened it in PowerPoint.

User · 09:03

These are Federico's prompts to be put on the material/layup/geometry slides. He gave them to me as plain text instead of screenshots, so please format them into the text bubbles so they look like my screenshots.

Material properties: I need a librarian agent to help me create a catalog of material properties for fiber reinforced composite laminas. We will focus on material properties of the UD lamina for now, not of the individual fibers or matrix. The catalog should have the following structure: Column 1: lamina ID number Column 2: name of fiber like T700 or AS-4 Column 3: name of polymer system like PEEK, DGEBA epoxy, ... Column 4: E1 (GPa) Column 5: E2 (GPa) Column 6: G12 (GPa) Column 7: nu12 Column 8: nu23 Column 9: G23 (GPa) Column 10: longitudinal strength under uniaxial compression (MPa) Column 11: longitudinal strength under uniaxial tension (MPa) Column 12: transverse strength under uniaxial compression (MPa) Column 13: transverse strength under uniaxial tension (MPa) Column 14: strength under shear (MPa) Column 15: ply thickness (mm) Column 16: density (kg/m^3) Column 17: the source link for the values provided The stiffnesses and poisson ratios should be sufficient to then be used by the tool laminate-properties in D:\Users\fbenazzo\projects\BenazzoVenkatadri_ThinShellSims.claude\skills. Instead, the strengths should be sufficient to satisfy the in-plane Tsai-Hill failure strength criterion. Rules:

1. The values in the catalog should be taken from referenced literature online, i.e. not wikipedia or unknown websites, but mostly properties actually measured by research groups or companies that used these materials
2. If nu23 or G23 are missing, you can use sensible estimates for them, but do report that their estimates with a * next to their value

## Find between 25 and 50 lamina properties and save the catalog as two .txt files (one with the units I asked for me to review and more easily read, and one with SI standard units, so Pa, m, N)

Layup catalog I want you to create another catalog which will be accessed by the main agent orchestrator to pick a layup for the next simulation. The catalog holds a list of layups to be used in the simulations. This should be layups that are:

- From 1 to 5 plies
- Symmetric
- Ply angles 0, 30, 45, 60, 90, -30, -45, -60 degrees

---

Geometry catalog I want you to create a catalog of the possible combinations of geometrical parameters for coilable booms. Let me explain, there is 4 types of booms to be considered: tape springs, omega or lenticular booms, TRAC booms, and corrugated booms (which are basically omega booms but the flange is allowed to have more than 3 arcs). I want you to look up in the literature the radii and angles and web-lengths used in different research projects and report these back in a txt catalog where you ID the type of coilable boom among the 4 above, the number of arcs, the radius of each arc, the subtended angle of each arc, and the length of the web. Make sure the following: 1) I want about 10-20 different geometries for each type 2) For each geometry, check that it can flatten (that is the arc-lengths of the flange between the webs are equal) 3) Feel free to offer some asymmetric geometries, that still satisfy point 2 4) Show me a few of these geometries by plotting them

**[20:57] Context compacted.** The session continued from this automatic summary:

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.

Summary: 1. Primary Request and Intent: - Build a slide deck in the same template and formatting as the final 11-11-26 group meeting deck (`11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx`). - Slide 1: title slide with the project title, the names (Federico Benazzo and Tara Venkatadri), "Lab Report 3 Demonstration of Agent System", and the AI disclaimer used on 11-11. - Slide 2: the full 11/4 system diagram, with a title saying all modules have been built and tested. - Then demo slides for the catalogs, the OpenSeesPy write/run pipeline, the Abaqus verification loop, and the full end-to-end baseline run. - Demo slides should roughly follow the template screenshot (diagram section, "What this section does", input/output examples, status Working/Failing). - Follow `Lab Report 3/Demo Slides Outline.docx`. Catalog prompt screenshots stay as placeholders. - Revision round 1: - Slide 3: each reference on one line; mention the permissions split (librarian is the only agent with web access). - Slide 4: change h to t. - Slide 5: no overlapping cross-section images. - Slide 6: the objective/constraints calculator does Hashin failure on the specified load cases plus flattening for coiling. - Slide 7: say the 1e-4 Hashin index is for bending. - Slide 8: mention the CPU limit. - Slide 10: L001 is on slide 3; "should fail" options should trigger material failure; "heavier" works against the lightest-boom objective. - Slide 11: explain strength ratio. - Slide 13: explain the coiling and stiffness rejections; next step says the orchestrator pre-screens and runs the grid search faster than the user. - Slide 14: remove the 11/10 date; "Output" instead of "Input/Output Example"; remove the "Failure case" row; say continuous vs. discrete geometry. - Throughout: use subscripts and remove personal pronouns. - Revision round 2: update the slide 2 system diagram to remove the 2 purple file-moving blocks and the ODB extraction step, because Claude now runs on the server with Abaqus. - Revision round 3: place the yellow box symmetrically between the translator and the verification (results comparison) agent. - Revision round 4 (latest): remove ", by Claude" from the yellow box.

2. Key Technical Concepts: - Deck building:
    - python-pptx builds the deck on top of the 11-11 deck: all of its slides are dropped and its masters and theme kept.
    - The 11-11 style: 13.33x7.5 in slides; grey F2F2F2 header bar with an orange ED7D31 rule; Calibri 36 pt bold titles; slide number at 12 pt; Caltech logo at the bottom right.
    - Styling details: tables with orange headers in white bold; peach FBE5D6 callouts; body text at least 18 pt; references at 12 pt.
    - Subscripts: `_{...}` markup in `_runs` becomes a subscript run (baseline -25000).
    - QA:
    - A custom PIL renderer (`render.py` in the scratchpad) stands in for LibreOffice or PowerPoint, neither of which is installed. It has known quirks: subscripts are drawn raised, and bullets are drawn on wrapped lines.
    - The pptx skill's `validate.py` runs with `--original` set to the 11-11 deck.
    - Diagram: matplotlib script `system_diagram_lab3.py`, run with the shells env `C:\Users\tvenkata\miniforge3\envs\shells\python.exe`.
    - Python environments:
    - Scratchpad venv with python-pptx, python-docx, pymupdf and Pillow: `C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-`

3c5983a9a11f/scratchpad/venv/Scripts/python.exe.
    - Skill scripts directory: C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts/.
3. Files and Code Sections: - Lab Report 3/build_demo_deck.py (created; the main build script). Key points:
    - DIAGRAM = os.path.join(HERE, "system_diagram_lab3.png").
    - diagram_crops() makes the crops:
    - The full image is trimmed of its empty bottom margin.
    - Crops of the green boxes (mat/lay/geo) with an orange highlight, top band (118,50,598,198).
    - bl crop (0,299,362,545).
    - br crop (350,297,722,528) with a white mask over the stray italic text on its left edge.
    - Helpers: add_text, add_box, add_pic, add_table (accepts a list for row_h, plus header=False), status_pill (h=0.45), header, label, what_it_does, refs(y=). _runs handles the SUB regex for subscripts.
    - There are 14 slides: 1. Title. 2. Diagram at w=7.3, y=0.95, with a checklist panel on the right. 3. Materials catalog. References are one line each at ref_y=6.6. 4. Layups catalog, with a t (mm) column. 5. Cross-section catalog. Drawings sit side by side with a 0.2 in gap, scaled to fit 4.5 in. 6. OpenSeesPy pipeline: prompt plus the deflection/rotation table. 7. Plots and three tiles (the Hashin tile says "in bending"). 8. Abaqus loop: prompt plus a CPU-limit callout. 9. Abaqus results table and images. 10. Sandboxed catalogs. 11. Tools-only baseline, with the strength ratio R = 1/√F explained. 12. Orchestrator run vs. baseline vs. Abaqus. 13. 27-tile pruning grid, with the C/S legend explanations and the next-step box. 14. Failure case (Palmeri and Laurenzi), with a headerless red table.
    - Speaker notes are on every slide. The notes for slides 2 and 8 describe the diagram update.
    - Lab Report 3/system_diagram_lab3.py (copy of Code for Project/11-4-26 (Tara)/system_diagram.py; the original is untouched).
    - The verification section now reads: python box(114, 19, 143, 30, 'exec', text='Abaqus run on the\nlab server\n(at most 8 CPUs)', fs=FS - 0.5) arrow([(109.5, 40), (109.5, 24.5), (114, 24.5)]) ... arrow([(143, 24.5), (147.5, 24.5), (147.5, 41)], 'ODB', (147.5, 33), ha='center') arrow([(59, 35.5), (130, 35.5), (130, 41)], 'results', (91, 35.5), ha='center', halo=True)
    - The purple "Script moved..." and "ODB extract moved..." boxes are removed.
    - Output goes to system_diagram_lab3.png.
    - Output files: Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx and Lab Report 3/system_diagram_lab3.png.
    - Inputs used:
    - The outline docx and its 21 images.
    - Catalogs in .claude/memory.*.
    - Code for Project/11-11-26 (Tara)/SUMMARY.md and ts_toy_minmass_REPORT.pdf (19 of 27 designs rejected, 8 iterations, optimum found at iteration 1).
    - The Palmeri and Laurenzi PDF: Biomimetics 2024, 9(8), 494.
4. Errors and fixes: - No soffice, PowerPoint or node available: installed python-pptx in a scratchpad venv and wrote a PIL renderer for QA. - Inline bash heredoc Python edits failed: once with "unexpected EOF" quoting errors, and once because \\n was mangled. Fix: write the edit scripts with the Write tool (edits2.py, e3.py, e4.py) and run them. - A sed edit for ", by Claude" didn't match because of escaping. Fixed with the Edit tool. - Layout problems found in renders and fixed:
    - Table wrapping on slides 3, 6 and 11.
    - Slide 11 table overlapping the logo.
    - Overlapping drawings on slide 5.
    - The trimmed diagram on slide 2 overlapping the right panel; fixed by fitting it by width (7.3 in).
    - The ∝ glyph, which Calibri lacks; replaced with "scales as".
5. Problem Solving: - All requested edits are applied. - The validator passes after every rebuild. - Remaining limitation: nothing has been checked in real PowerPoint. Subscripts in particular should be checked there.
6. All user messages: - Initial request: "Build a slide deck with the same template and formatting requirements as the final iteration of the 11-11-26 group meeting presentation. The 1st slide should be a title slide with the title of our project, our names, and "Lab Report 3 Demonstration of Agent System", and an AI usage disclaimer like the 11-11 group meeting one. The 2nd slide should be the full system diagram from 11/4 with all text (it is ok if it is smaller than the required font sizes from the 11/11 presentation). In the slide title, say all modules have been built and tested. Then there should be slides showing demonstrations of our components: the catalogs, the OpenSeesPy write/run pipeline, the Abaqus verification loop, and the full end-to-end baseline run. These should roughly follow the screenshot of the slide above. Use the outline Demo Slides Outline in F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3" (with a template screenshot). - Revision list (full text): "Slide 3: make sure each reference fits on 1 line so the references don't bleed off the bottom of the slide / Slide 3: Mention permissions split for librarian to be a separate agent from orchestrator (only agent with

web access) / Slide 4: Replace ply thickness h with t / Slide 5: Make sure there's no overlap in the images of different cross-sections / Slide 6: Say that objective/constraints calculator does Hashin failure on specified loading cases and also flattening for coiling to match deployable structures application of this project / Slide 7: Say the 1e-4 Hashin index is for bending to distinguish it from the coiling one / Slide 8: Mention the limit on number of CPUs used on shared lab folder / Slide 10: Mention that L001 is on slide 3. Mention that the "should fail" ones should trigger material failure, and that heavier is against the objective of having the lightest boom / Slide 11: Explain what strength ratio means / Slide 13: More explanation as to why some designs were rejected for coiling or stiffness. In next step, add that optimization orchestrator can pre-screen for viable designs and conduct the necessary grid search much faster than the user / Slide 14: Delete 11/10 (the date) from "Input". Replace "Input-Output Example" with "Output". Get rid of the row that says "Failure case". Specify that it's continuous variation in geometry rather than discrete variation. / Throughout slides, do subscripts for things like sigma_x, u_z, etc, and remove personal pronouns" - "Update the system diagram on slide 2 so the 2 purple blocks where the user moves things from one machine to another are eliminated and ODB extraction is no longer necessary, since Claude is now on the server with Abaqus" - "Make the yellow box symmetrically between the translator and verification agent" - "Get rid of ", by Claude"" - Constraints carried over from the project's CLAUDE.md:

   - Run Abaqus only via run_abaqus.py, with at most 8 CPUs.
   - Never write or change anything outside the ME295 folder on the shared server.
   - Running Log entries record date, person and entry.
   - Organization instructions: protect sensitive data, and never share the user's email with external services.

7. Pending Tasks: - None explicitly pending. - Still placeholders: Federico's catalog prompt screenshots on slides 3-5, which the user is to paste in. - Nothing has been committed. Commit only if the user asks.
8. Current Work: - Removed ", by Claude" from the yellow Abaqus box in `Lab Report 3/system_diagram_lab3.py`. The text is now `'Abaqus run on the\nlab server\n(at most 8 CPUs)'`. - Regenerated the PNG with the shells python and rebuilt the deck. Validation passed, and the slide 8 render confirmed the change. - wrap_up was called with the recap. The closing text message to the user had not yet been written.
9. Optional Next Step: - Send the brief closing message: the yellow box now reads "Abaqus run on the lab server (at most 8 CPUs)"; the diagram and deck are regenerated, slides 2 and 8 updated, the validator passes, and it has not been checked in PowerPoint. - No further work unless the user asks.

If you need specific details from before compaction (like exact code snippets, error messages, or content you generated), read the full transcript at: C:\Users\tvenkata.claude\projects\F--Tara-BenazzoVenkatadri-ThinShellSims\07583a8d-5e34-4ade-abc5-3c5983a9a11f.jsonl
Continue the conversation from where it left off without asking the user any further questions. Resume directly — do not acknowledge the summary, do not recap what was happening, do not preface with "I'll continue" or similar. Pick up the last task as if the break never happened.

The messages after this summary are the most recent messages from before compaction, kept verbatim. The summary was written without seeing them, so something it says has not happened yet may already have happened in them.

**update_plan** task list

```
[in_progress] Find placeholders and prompt bubble style
[pending] Build text bubbles for slides 3-5
[pending] Rebuild, validate and render
```

```
Plan shown (3 steps).
```

**Grep**

```
{
 "pattern": "placeholder|Placeholder|prompt|def add_pic|def label|def slide_[345]|^# ---",
 "path": "F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Lab Report 3\\build_demo_deck.py",
 "output_mode": "content",
 "-i": true
}
```

```
51:# ---------------------------------------------------------------------------- assets
107:# ---------------------------------------------------------------------------- helpers
211:def add_pic(slide, blob, x, y, w=None, h=None, border=True, name=None):
294:def label(slide, x, y, w, text, size=18):
306:# ---------------------------------------------------------------------------- deck
351:                  "The next slides demonstrate each part with the prompt it received and the output it produced.")
377:        label(s, X, 1.3, W, "Input: Federico's prompt to the librarian agent")
379:                [[("[Placeholder: screenshot of Federico's prompt that created this catalog]",
404:        "PLACEHOLDER: paste the screenshot of the prompt that created the material catalog.\n"
431:        "PLACEHOLDER: paste the screenshot of the prompt that created the layup catalog.\n"
[... shortened ...]
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
107     # ---------------------------------------------------------------------------- helpers
108
109     SUB = re.compile(r"_\{([^}]*)\}")
110
111
112     def _runs(p, runs, size, bold, color, italic=False):
113         """Add runs; "_{...}" inside a run's text is written as a subscript run."""
114         if isinstance(runs, str):
115             runs = [(runs, {})]
116         split = []
[... shortened ...]
```

**Bash** Show asset loading code and scratchpad files

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -n 51,106p build_demo_deck.py; ls
"C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/" | head
-50
```

```
# ---------------------------------------------------------------------------- assets

def outline_images():
    """Screenshots of the outline .docx, in document order (img01 ... img21)."""
    d = docx.Document(OUTLINE)
    out = []
    for p in d.paragraphs:
        for b in p._p.findall(".//" + dqn("a:blip")):
            out.append(d.part.related_parts[b.get(dqn("r:embed"))].blob)
    return {f"img{i + 1:02d}": blob for i, blob in enumerate(out)}
[... shortened ...]
```

**Bash** List extracted outline images and renders

```
cd "C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/" && ls
docimg img r | head -60
```

```
docimg:
img01.png
img02.png
img03.png
img04.png
img05.png
img06.png
img07.png
img08.png
img09.png
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/docimg/img13.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Claude · 09:04**

Federico's prompts are long, so I'm laying them out as grey prompt bubbles in the style of your screenshots. Next I'll fit them on slides 3-5.

**Write** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\prompt_bubbles.py

```
"""
Federico's librarian prompts (given as plain text) drawn as chat-style prompt bubbles that match the
screenshots in "Demo Slides Outline.docx": light grey rounded box, Segoe UI, bullets with hanging indent.
Used by build_demo_deck.py for slides 3-5. Text is verbatim; only the layout (bullets, list columns) is set here.

  python prompt_bubbles.py      writes bubble_material.png, bubble_layup.png, bubble_geometry.png for a look
"""
import io
```

```
import os

from PIL import Image, ImageDraw, ImageFont

[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\prompt_bubbles.py (file state is current
in your context — no need to Read it back)
```

**Bash** Render prompt bubbles and report heights

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" &&
V="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad/venv/Scripts/python.exe"; "$V" prompt_bubbles.py
```

```
material 4.486666666666666 in tall
layup 1.1133333333333333 in tall
geometry 2.0 in tall
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\bubble_material.png

```
[image]
```

**Bash** Find hard-coded slide number references

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n -i "slide [0-9]\|slides [0-9]\|\"3-5\"\|\"6-7\"\|\"8-
9\"\|\"10-13\"\|\"14\"" build_demo_deck.py
```

```
355:    items = [("Catalogs", "librarian agent", "3-5"),
356:             ("OpenSeesPy write/run pipeline", "tools", "6-7"),
357:             ("Abaqus verification loop", "comparison agent", "8-9"),
358:             ("End-to-end baseline", "orchestrator agent", "10-13"),
359:             ("Failure case", "omega boom", "14", )]
621:               ["Should be optimal", "L001 (slide 3)", "[0/0/0]", "R = 20 mm, 180°"],
627:             [["L001: ", {"bold": True}), ("IM7/8552 row of the material catalog (slide 3)", {})],
822:               ["Outcome", "Abandoned for the simpler tape-spring baseline (slides 10-13)"]],
```

**Bash** Inspect material catalog files and header

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims" && ls .claude/memory.material_property_catalog/ && head -30
.claude/memory.material_property_catalog/UD_lamina_catalog_readable.txt | cut -c1-200
```

```
AIAA_SciTech_2025_paper.pdf
UD_lamina_catalog_SI.txt
UD_lamina_catalog_readable.txt
lamina_catalog_to_SI.py
# UD lamina property catalog (readable units: GPa, MPa, mm)
# Maintained by the CompositeLaminaLibrarian agent. SI twin: UD_lamina_catalog_SI.txt
# (generated from this file by lamina_catalog_to_SI.py; do not edit the SI file by hand).
#
# Columns: ID | Fiber | Polymer system | E1 | E2 | G12 | nu12 | nu23 | G23 | Xc | Xt | Yc | Yt | S12 | t_ply | rho | Source
#   Xc/Xt = longitudinal compressive/tensile strength, Yc/Yt = transverse compressive/tensile strength,
[... shortened ...]
```

**Edit** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\prompt_bubbles.py

```
-     ("gap",),
    ("p", "Find between
+     ("p", "Find between
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
The file F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\prompt_bubbles.py has been updated successfully. (file state
is current in your context — no need to Read it back)
```

```
1       """Build the Lab Report 3 demonstration deck in the style of the 11-11-26 group meeting deck.
2
3       Run with any Python that has python-pptx, python-docx and Pillow:
4           python "Lab Report 3/build_demo_deck.py"
5       Inputs: the 11-11-26 deck (theme, title background, Caltech logo, section drawings), the screenshots
6       in "Lab Report 3/Demo Slides Outline.docx" and the system diagram made by system_diagram_lab3.py.
7       Output: "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx".
8       """
9       import io
```

```
10      import os
[... shortened ...]
```

Claude · 09:05

Now the edits to the build script.

**Write** C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-3c5983a9a11f\scratchpad\e5.py

```
import sys
f = sys.argv[1]
s = open(f, encoding="utf8").read()


def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)


rep('''in "Lab Report 3/Demo Slides Outline.docx" and the system diagram made by system_diagram_lab3.py.''',
[... shortened ...]
```

```
File created successfully at: C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\e5.py (file state is current in your context — no need to Read it back)
```

**Edit** C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-3c5983a9a11f\scratchpad\e5.py

```
- rep('''from pptx.util import Emu, Inches, Pt
''', '''from pptx.util import Emu, Inches, Pt

import sys
sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
''')

+
```

```
The file C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\e5.py has been updated successfully. (file state is current in your context — no need to Read it
back)
```

**Bash** Apply edits, rebuild deck, render for QA

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims" &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" "$SP/e5.py" "Lab Report 3/build_demo_deck.py" && "$V" "Lab Report
3/build_demo_deck.py" && "$V" "$SP/render.py" "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s"; rm -
f "Lab Report 3"/bubble_*.png
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.22 in past set height, bottom at 6.88 in
slide 10: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
Traceback (most recent call last):
  File "C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\render.py", line 200, in <module>
    print(i)
    ~~~~~^^^
[... shortened ...]
```

Claude · 09:06

Okay, I'll make the column list one continuous list, as in a Claude prompt. That makes the material bubble taller, so first I'll check what font size still fits slide 3.

**Bash** Set one column, measure material bubble heights

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/             "Column 17: the source link for the
values provided"\], 2),/             "Column 17: the source link for the values provided"], 1),/' prompt_bubbles.py && grep
```

```
-n '], 1),' prompt_bubbles.py && V="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad/venv/Scripts/python.exe"; "$V" -c "
import io, prompt_bubbles as pb
from PIL import Image
for pt in (8,8.5,9,9.5):
  for w in (7.88,):
    print(pt, Image.open(io.BytesIO(pb.bubble(pb.MATERIAL,w,pt))).size[1]/300)
"
```

```
39:              "Column 17: the source link for the values provided"], 1),
8 5.09
8.5 5.41
9 5.726666666666667
9.5 6.42
[This command modified 1 file you've previously read: prompt_bubbles.py. Call Read before editing.]
```

**Bash** Locate prompt layout lines in build script

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n "y = 1.3$\|if prompt:\|label(s, X, y, W,
\"Input\|pb.bubble\|y += 0.42 + bh\|io_label=\"Input\"" build_demo_deck.py
```

```
379:        y = 1.3
380:        if prompt:
381:            label(s, X, y, W, "Input: Federico's prompt to the librarian agent")
382:            _, _, bh = add_pic(s, pb.bubble(prompt, W, 9.5), X, y + 0.42, w=W, border=False,
384:            y += 0.42 + bh + 0.15
404:        prompt=pb.MATERIAL, io_label="Input")
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
370         # 3-5 ------------------------------------------------------------- catalogs
371         def catalog_slide(title, crop, desc, out_label, cols, rows, notes, extra=None, ref=None, row_h=0.45,
372                           ref_y=6.93, prompt=None, io_label="Input / Output Example"):
373             s = new_slide(title, notes)
374             _, _, h = add_pic(s, crop, 0.3, 0.85, w=4.5, name="diagram_section")
375             what_it_does(s, 0.3, 0.85 + h + 0.2, 4.5, 3.0, desc)
376             X, W = 5.15, 7.88
377             label(s, X, 0.78, 5.5, io_label, size=20)
378             status_pill(s, X + W - 1.75, 0.78)
379             y = 1.3
[... shortened ...]
```

**Bash** Apply layout edits, rebuild and render

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && cat >
/c/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e6.py
<<'EOF'
import sys
f = sys.argv[1]
s = open(f, encoding="utf8").read()
def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)
rep('''ref_y=6.93, prompt=None, io_label="Input / Output Example"):''',
    '''ref_y=6.93, prompt=None, io_label="Input / Output Example", prompt_pt=9.5):''')
rep('''        y = 1.3
        if prompt:
            label(s, X, y, W, "Input: Federico's prompt to the librarian agent")
            _, _, bh = add_pic(s, pb.bubble(prompt, W, 9.5), X, y + 0.42, w=W, border=False,
                               name="prompt_librarian")
            y += 0.42 + bh + 0.15''',
    '''        y = 1.3
        if prompt:
[... shortened ...]
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 3: text overflow 0.31 in in 'TextBox 9': "Input: Federico's prompt to the librarian agent"
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.22 in past set height, bottom at 6.88 in
```

```
slide 10: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
slide 12: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
slide 15: table 'Table 10' grows 1.06 in past set height, bottom at 4.76 in
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-04.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-05.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Bash** Apply fixes for label, table widths, font size

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && cat >
/c/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e7.py
<<'EOF'
import sys
f = sys.argv[1]
s = open(f, encoding="utf8").read()
def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)
rep('''io_label="Input / Output Example", prompt_pt=9.5):''', '''io_label="Input / Output Example", prompt_pt=8.5):''')
rep('''        label(s, X, 0.78, 5.5, io_label, size=20)''', '''        label(s, X, 0.78, 6.0, io_label, size=20 if rows
else 18)''')
rep(''', prompt_pt=8.5)''', ''')''')
rep('''        "Output: sample rows of the material catalog",''', '''        "Sample rows of the material catalog",''')
rep('''        [0.8, 1.75, 2.45, 1.48, 1.4],''', '''        [0.8, 1.65, 2.75, 1.48, 1.2],''')
rep('''        extra=geo_drawings, prompt=pb.GEOMETRY,''', '''        extra=geo_drawings, prompt=pb.GEOMETRY,''')
open(f, "w", encoding="utf8").write(s)
print("ok")
EOF
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" "$SP/e7.py" build_demo_deck.py && grep -n "row_h=0.45)" build_demo_deck.py
```

```
Traceback (most recent call last):
  File "C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\e7.py", line 10, in <module>
    rep(''', prompt_pt=8.5)''', ''')''')
    ~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\tvenkata\AppData\Local\Temp\16\claude\91fa69c9debc8176\07583a8d-5e34-4ade-abc5-
3c5983a9a11f\scratchpad\e7.py", line 6, in rep
    assert s.count(a) == 1, a[:70]
           ^^^^^^^^^^^^^^^
AssertionError: , prompt_pt=8.5)
```

**Bash** Fix edit script and apply

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; sed -i
"s/^rep(''', prompt_pt=8.5)''', ''')''')$/rep('''librarian agent\", prompt_pt=8.5)''', '''librarian agent\")''')/"
"$SP/e7.py" && V="$SP/venv/Scripts/python.exe"; "$V" "$SP/e7.py" build_demo_deck.py && grep -n "row_h=0.45)"
build_demo_deck.py
```

```
ok
486:        row_h=0.45)
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**Bash** Set geometry table row height, rebuild, render

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -n 480,487p build_demo_deck.py && sed -i
'486s/row_h=0.45)/row_h=0.42)/' build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s"
```

```
"black = bonded web.",
        extra=geo_drawings, prompt=pb.GEOMETRY,
        ref=["[5] Benazzo, F., & Pellegrino, S. (2025). Design and manufacturing of ultra-thin thermoplastic "
             "composites for coilable longerons. AIAA SciTech 2025 Forum.",
             "[6] Fernandez, J. M., & Volle, C. E. (2021). Corrugated rollable tubular booms. AIAA SciTech 2021 "
             "Forum."],
        row_h=0.45)
    # the geometry description is short so the drawings fit under it (left column, y = 5.25)
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Claude · 09:08**

Slides 3 and 6 now fit. One small fix remains: "Corrugated [6]" wraps on slide 6, so I'll rebalance the column widths and then run the validator.

**Bash** Adjust widths, rebuild, render check, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/        \[0.8, 1.65, 2.75, 1.48, 1.2\],/        [0.8,
1.75, 2.75, 1.38, 1.2],/' build_demo_deck.py && grep -c "1.75, 2.75, 1.38" build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide [3-6]:"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-
agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --
original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -3
```

```
1
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.35 in
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
Federico's three prompts are now grey Claude-style prompt bubbles on the catalog slides. The material prompt needed its own
slide, so the deck has 15 slides and the slide-number references were updated. The validator passes; the deck has not been
checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

**Claude · 09:08**

Federico's three prompts are now on the catalog slides as grey prompt bubbles like your screenshots. The column list is one continuous list, as you asked. The deck rebuilds and passes the file validator, but I haven't opened it in PowerPoint.

**The material prompt needed its own slide.** As one continuous list it is about 5.4 in tall, which leaves no room for the output table on the same slide. The material catalog is now two slides, so the deck has 15: - **Slide 3, "Catalogs: Material Properties (Input)":** the diagram section and "What this section does" on the left, Federico's full prompt on the right. - **Slide 4, "Catalogs: Material Properties (Output)":** the sample-row table and the references. The left side now says what came back: 38 laminas, within the 25-50 he asked for, saved as two .txt files (readable units and SI), with * marking estimates.

Every slide after that moved up by one. I updated all the places that name a slide number: the slide 2 checklist (now 3-6, 7-8, 9-10, 11-14, 15), "L001 (slide 4)" on the sandboxed-catalogs slide, and "slides 11-14" on the failure-case slide.

**Layup and geometry prompts (slides 5 and 6):** each bubble replaces the placeholder box above its table. On slide 6 I made the table columns a bit narrower so the table no longer runs into the Caltech logo.

**Details:** - **Text size:** the bubble text is 8.5 pt in all three bubbles. That is close to the size of the other screenshots in the deck. - **Wording:** Federico's text is word for word. The only formatting changes are bullets for the layup list and hanging indents for the numbered rules. - **Editing:** the bubbles are generated as images by the new prompt_bubbles.py, and the prompt text lives in that file. To change the wording, edit it there and rerun build_demo_deck.py.

User · 09:11

What are you still doing?

Claude · 09:11

Nothing. The prompt bubbles were the last thing I worked on, and that change is done. The deck rebuilt to 15 slides and passes the file validator.

The deck hasn't been opened in PowerPoint yet, and nothing has been committed. Tell me if you want either of those, or any more changes to the slides.

User · 09:15

Get rid of where it says "Federico's prompt to the librarian agent"

**Bash** Find the label occurrences

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n "Federico's prompt to the librarian agent"
build_demo_deck.py && sed -n 379,392p build_demo_deck.py
```

```
382:                label(s, X, y, W, "Input: Federico's prompt to the librarian agent")
406:        "Federico's prompt to the librarian agent, verbatim (laid out as a prompt bubble like the other "
408:        prompt=pb.MATERIAL, io_label="Input: Federico's prompt to the librarian agent")
        y = 1.3
        if prompt:
            if rows:
                label(s, X, y, W, "Input: Federico's prompt to the librarian agent")
                y += 0.42
            else:
                y = 1.27  # prompt-only slide: the bubble starts right under the "Input" label
[... shortened ...]
```

**Bash** Replace labels with "Input", rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i '382s/"Input: Federico'"'"'s prompt to the librarian
agent"/"Input"/; 408s/io_label="Input: Federico'"'"'s prompt to the librarian agent"/io_label="Input"/' build_demo_deck.py
&& grep -n "Federico's prompt to the librarian agent\"" build_demo_deck.py;
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide [3-6]:"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-
agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-
5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --
original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.35 in
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-05.png

```
[image]
```

**wrap_up** wrap-up

```
"Federico's prompt to the librarian agent" is gone from slides 3, 5 and 6; the labels now just say "Input". The deck is rebuilt and the validator passes. Nothing is committed and it hasn't been opened in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

**Claude · 09:15**

I removed "Federico's prompt to the librarian agent" from slides 3, 5 and 6. Each label now just says "Input", to match "Output" on slide 4 and the "Output: sample rows…" labels on slides 5 and 6.

I rebuilt the deck and it passes the file validator. The speaker notes for slide 3 still mention that the prompt is Federico's; they aren't visible on the slide.

**User · 09:15**

And put boxes around the bubbles like in the later slides

**Bash** Locate bubble drawing and placement lines

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n "im = Image.new\|rounded_rectangle\|H = int\|d.text((x" prompt_bubbles.py && grep -n 'name="prompt_librarian"' -B1 build_demo_deck.py
```

```
131:    H = int(round(y + pad_y))
132:    im = Image.new("RGBA", (W, H), (255, 255, 255, 0))
134:    d.rounded_rectangle([0, 0, W - 1, H - 1], radius=int(0.9 * font_pt * px), fill=BG)
137:        d.text((x, yy + (lh - f.size * 1.33) / 2 + 0.1 * asc), t, font=f, fill=FG)
386-            _, _, bh = add_pic(s, pb.bubble(prompt, W, prompt_pt), X, y, w=W, border=False,
387:                               name="prompt_librarian")
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\prompt_bubbles.py

```
84                  lines.append(cur)
85                  cur = w
86              else:
87                  cur = t
88          lines.append(cur)
89          return lines
90
91
92      def bubble(items, width_in, font_pt, dpi=300):
93          """PNG bytes of the bubble, width_in wide, text at font_pt when placed at that width."""
[... shortened ...]
```

**Bash** Add white margin and border to bubbles, rebuild

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && cat > /c/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e8.py <<'EOF'
import sys
def edit(f, pairs):
    s = open(f, encoding="utf8").read()
    for a, b in pairs:
        assert s.count(a) == 1, a[:70]
        s = s.replace(a, b)
    open(f, "w", encoding="utf8").write(s)
edit("prompt_bubbles.py", [
 ('''def bubble(items, width_in, font_pt, dpi=300):
    """PNG bytes of the bubble, width_in wide, text at font_pt when placed at that width."""
    px = dpi / 72.0
    f = ImageFont.truetype(FONTS + "segoeui.ttf", int(round(font_pt * px)))
    W = int(round(width_in * dpi))''',
  '''def bubble(items, width_in, font_pt, dpi=300, margin_in=0.1):
    """PNG bytes of the bubble on a white margin (like the screenshots), width_in wide in total,
    text at font_pt when placed at that width."""
    px = dpi / 72.0
[... shortened ...]
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
```

```
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.55 in
All validations PASSED!
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-06.png

```
[image]
```

**Claude · 09:16**

The bubbles now have the same thin grey frame and white margin as the later screenshots. On slide 3 the framed bubble now touches the Caltech logo, so I'll pull it up slightly.

**Bash** Tighten margin, raise slide 3 bubble, check bottoms

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/dpi=300, margin_in=0.1):/dpi=300, margin_in=0.08):/'
prompt_bubbles.py && sed -i 's/                y = 1.27  # prompt-only slide/                y = 1.2  # prompt-only slide/'
build_demo_deck.py && grep -c "margin_in=0.08" prompt_bubbles.py && grep -c "y = 1.2  # prompt-only" build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && "$V" -c "
from pptx import Presentation
p=Presentation('Lab Report 3 Demonstration of Agent System.pptx')
for i in (2,4,5):
  for sh in p.slides[i].shapes:
    if sh.name=='prompt_librarian': print(i+1, round(sh.top/914400,2), round((sh.top+sh.height)/914400,2))
" && SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-
43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3
Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite
Booms (11-11-26).pptx" 2>&1 | tail -1; PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3 Demonstration of Agent
System.pptx" "$SP/r/s" | grep "slide [3-6]:"
```

```
1
1
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
3 1.2 6.77
5 1.72 2.93
6 1.72 3.6
All validations PASSED!
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.51 in
[This command modified 2 files you've previously read: build_demo_deck.py, prompt_bubbles.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
The prompt bubbles on slides 3, 5 and 6 now sit inside a thin grey frame with a white margin, like the later screenshot
slides. The slide 3 bubble was moved up so it clears the Caltech logo. The validator passes; not checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

**Claude · 09:16**

The prompt bubbles on slides 3, 5 and 6 now have the same thin grey frame and white margin as the screenshots on the later slides.

On slide 3 the framed bubble had started to touch the Caltech logo. I moved it up slightly and narrowed the margin, so it now ends at 6.77 in, just above the logo at 6.82 in. I rebuilt the deck and it passes the file validator; I haven't checked it in PowerPoint.

**User · 09:16**

And get rid of (Input) and (Output) in the slide 3/4 titles

**Bash** Drop (Input)/(Output) from titles, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/"Catalogs: Material Properties (Input)"/"Catalogs: Material Properties"/; s/"Catalogs: Material Properties (Output)"/"Catalogs: Material Properties"/' build_demo_deck.py && grep -c '"Catalogs: Material Properties"' build_demo_deck.py && SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
2
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
Slides 3 and 4 are now both titled "Catalogs: Material Properties". The deck is rebuilt and the validator passes; nothing is committed and it hasn't been opened in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

Claude · 09:17

I removed "(Input)" and "(Output)", so slides 3 and 4 are both titled "Catalogs: Material Properties". The slide labels still show which slide is which: slide 3 is labeled "Input" and slide 4 "Output".

I rebuilt [the deck] and it passes the file validator.

User · 09:28

Move the box with the screenshot down on slide 3 so it doesn't overlap the "Working" bubble Slide 4: Remove the * = estimate from the bottom of the slide because it's already in the "what this section does" Slide 9: crop the screenshot of the system diagram so the bottom of the boxes above the verification loop boxes are not visible Slide 10: For the 4 figures, label OpenSees and Abaqus, and u_z and sigma_xx (with subscripts), in 18-pt font so it's clear what each figure represents (but keep the little descriptions in the top left of each figure since they have more detail) For all slides where the input is a prompt, say "Input: prompt to ..." where ... describes what the prompt does in a few words

**update_plan** task list

```
[in_progress] Review slides 3, 4, 9, 10 and prompt labels
[pending] Fix slide 3 bubble and slide 4 footnote
[pending] Crop slide 9 diagram section
[pending] Label the four slide 10 figures
[pending] Reword all prompt input labels
[pending] Rebuild, render and validate
```

```
Plan shown (6 steps).
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-09.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

```
[image]
```

**Bash** Find input labels and slide section markers

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n '"Input\|Input:' build_demo_deck.py; grep -n "# 9 \|# 10 \|# 8 \|# 9-\|-- Abaqus" build_demo_deck.py
```

```
372:                     ref_y=6.93, prompt=None, io_label="Input / Output Example", prompt_pt=8.5):
382:                label(s, X, y, W, "Input")
385:                y = 1.2  # prompt-only slide: the bubble starts right under the "Input" label
408:        prompt=pb.MATERIAL, io_label="Input")
506:    label(s, X, 0.78, 5.5, "Input / Output Example", size=20)
508:    label(s, X, 1.25, W, "Input: prompt for one tape spring under a 1 N tip load")
569:    label(s, X, 0.78, 5.5, "Input / Output Example", size=20)
```

```
571:    label(s, X, 1.25, W, "Input: prompt (same tape spring as the previous slides)")
630:    label(s, 0.3, 1.6, 5.0, "Input: prompt to create the toy catalogs")
677:    label(s, X, 0.78, W, "Input: prompt to run the tools on the known optimum")
[... shortened ...]
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
370         # 3-5 ------------------------------------------------------------- catalogs
371         def catalog_slide(title, crop, desc, out_label, cols, rows, notes, extra=None, ref=None, row_h=0.45,
372                           ref_y=6.93, prompt=None, io_label="Input / Output Example", prompt_pt=8.5):
373             s = new_slide(title, notes)
374             _, _, h = add_pic(s, crop, 0.3, 0.85, w=4.5, name="diagram_section")
375             what_it_does(s, 0.3, 0.85 + h + 0.2, 4.5, 3.0, desc)
376             X, W = 5.15, 7.88
377             label(s, X, 0.78, 6.0, io_label, size=20 if rows else 18)
378             status_pill(s, X + W - 1.75, 0.78)
379             y = 1.3
[... shortened ...]
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
585         # 9 ------------------------------------------------------------- Abaqus loop: results
586         s = new_slide("Abaqus Verification Loop: Results",
587                       "Suggested presenter: Tara.\n"
588                       "Comparison written by the OpenSeesAbaqusResultsComparator agent from the OpenSeesPy .npz and the
"
589                       "Abaqus ODB extract only. Pictures: deformation x20, grey = undeformed, black = clamped nodes;
top: "
590                       "u_z, bottom: sigma_xx on the top ply.\n"
591                       "Differences under 1% are acceptable: OpenSeesPy (ASDShellQ4 with LayeredShell sections) and "
592                       "Abaqus (S4R with a composite layup) use different shell elements, integrate the layers "
593                       "differently and apply the loads/BCs differently (rigid links vs. coupling).")
594         add_table(s, 0.3, 0.85, [2.5, 1.3, 1.3, 1.0],
[... shortened ...]
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
700                       "(baseline 2.5 mm, orchestrator 5 mm). Abaqus (5 mm mesh): tip deflection +2.6%, peak ply stress "
701                       "+2.1% (study report); the 42.6 MPa is the top ply, the 47.7 / 50.0 MPa are the peak over all
plies.\n"
702                       "Buckling (asked to stop once past 1 N): perfect-boom phase-B path reached 1037 N mm > 1000 N mm "
703                       "root moment with no limit point; Abaqus buckling on the perfect geometry: linear eigenvalue "
704                       "4336 N mm, Riks limit point 1395 N mm, both above 1000 N mm. Mesh convergence 5 -> 3 mm: tip "
705                       "stiffness 0.72%.")
706         label(s, 0.3, 0.78, 4.2, "Input: prompt to the orchestrator")
707         add_pic(s, shots["img19"], 0.3, 1.2, h=3.85, name="prompt_orchestrator")
708         add_box(s, 0.3, 5.25, 3.95, 1.45,
709                 [[("Optimum = human-chosen baseline: ", {"bold": True}), ("L001, [0/0/0], R = 20 mm", {})]],
[... shortened ...]
```

**Read** F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\build_demo_deck.py

```
815                       "Conversation 1).\n"
816                       "This was first considered as the baseline: reproduce the omega (collapsible tubular mast) "
817                       "geometry optimization of Palmeri and Laurenzi (2024) for maximum buckling load / moments. It "
818                       "turned out to be outside the scope of the optimization orchestrator: the material and layup were
"
819                       "given by the user instead of taken from the catalogs, the geometry (r1, alpha) was optimized "
820                       "continuously instead of picked from discrete catalog rows, and the buckling simulations differed
"
821                       "from the ones the existing skills set up. Claude worked for hours but could not even get a mesh "
822                       "convergence study to work on the paper's optimal design, because of OpenSeesPy's limited buckling
"
823                       "capabilities for this closed section. It was abandoned in favor of the simpler tape-spring "
824                       "baseline of the previous slides.")
[... shortened ...]
```

**Bash** Show slide 7 and 9 label context

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -n 500,512p build_demo_deck.py; sed -n 563,575p
build_demo_deck.py; grep -n "catalog_slide(\|prompt=pb\.\|product data sheet" build_demo_deck.py
```

```
what_it_does(s, 0.3, 0.85 + h + 0.12, 4.6, 3.0,
                "Writes the material/layup, mesh and load/BC code, joins it into one OpenSeesPy script and runs "
                "it. Computes the mass and the Hashin failure index for the specified load cases, plus flattening "
                "for coiling (the deployable-boom application), for the orchestrator. Draws the results."
                )
    X, W = 5.15, 7.88
    label(s, X, 0.78, 5.5, "Input / Output Example", size=20)
    status_pill(s, X + W - 1.75, 0.78)
    label(s, X, 1.25, W, "Input: prompt for one tape spring under a 1 N tip load")
    add_pic(s, shots["img03"], X, 1.67, w=5.9, name="prompt_opensees")
[... shortened ...]
```

**Bash** Apply all five edits, rebuild, render, check positions

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && cat >
/c/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e9.py
<<'EOF'
f = "build_demo_deck.py"
s = open(f, encoding="utf8").read()
def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)
# catalog slides: per-slide input label; slide 3 bubble below the Working pill
rep('''io_label="Input / Output Example", prompt_pt=8.5):''',
    '''io_label="Input / Output Example", prompt_pt=8.5, input_label="Input"):''')
rep('''                label(s, X, y, W, "Input")''', '''                label(s, X, y, W, input_label)''')
rep('''                y = 1.2  # prompt-only slide: the bubble starts right under the "Input" label''',
    '''                y = 1.33  # prompt-only slide: the bubble starts under the label and the status pill''')
rep('''        prompt=pb.MATERIAL, io_label="Input")''',
    '''        prompt=pb.MATERIAL, io_label="Input: prompt to build the material catalog", prompt_pt=8.25)''')
rep('''        row_h=0.42, prompt=pb.LAYUP)''',
    '''        row_h=0.42, prompt=pb.LAYUP, input_label="Input: prompt to build the layup catalog")''')
rep('''        extra=geo_drawings, prompt=pb.GEOMETRY,''',
[... shortened ...]
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 4: table 'Table 12' grows 0.42 in past set height, bottom at 4.40 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.51 in
slide 10: table 'Table 6' grows 0.23 in past set height, bottom at 3.78 in
slide 12: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
slide 13: text overflow 0.25 in in 'TextBox 6': 'Input: prompt to find the lightest boom'
slide 15: table 'Table 10' grows 1.06 in past set height, bottom at 4.76 in
s3 bubble 1.33 6.74
[... shortened ...]
```

**Claude · 09:29**

All five edits are applied. One new label, "Input: prompt to find the lightest boom" on slide 13, wraps onto a second line, so I'm shortening it and then checking slides 3, 9 and 10 visually.

**Bash** Shorten slide 13 label, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/"Input: prompt to find the lightest boom"/"Input:
prompt for the lightest boom"/' build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 13"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-
mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts";
"$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting
Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-09.png

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-03.png

```
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
```

```
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

```
[image]
```

```
[image]
```

```
[image]
```

**wrap_up** wrap-up

```
All five changes are in: the slide 3 bubble sits below the Working pill, slide 4 lost the duplicate "* = estimate", the
slide 9 diagram crop starts below the orchestrator box, the slide 10 figures are labeled, and every prompt label now says
what the prompt does. The validator passes; not checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Do not call wrap_up again this turn; the
advice that follows is for your next call. recap is two or three sentences (about 300 characters); this one ran 312. Keep it
to what is now true and where it stands. Your closing message is your answer to the user: write it now, as the last thing
you say this turn.
```

Claude · 09:30

I made all five changes. The deck passes the file validator, but I haven't opened it in PowerPoint.

- **Slide 3:** the prompt box now starts below the "Working" pill. To keep it clear of the Caltech logo, I shrank the bubble text slightly, from 8.5 pt to 8.25 pt.
- **Slide 4:** I removed "* = estimate" from the end of the last reference.
- **Slide 9:** the diagram crop now starts below the orchestrator box, so the bottom edges of the boxes above no longer show. Only the short arrow stubs leading up into them are still visible.
- **Slide 10:** the four figures have 18 pt bold labels: "OpenSees" and "Abaqus" over the columns, and $u_z$ and $\sigma_{xx}$ beside the rows, written as real subscripts. The small descriptions in each figure's top left are kept. The figures are a little smaller to make room for the labels.
- **Prompt labels:** each one now says what the prompt does:

| Slide | Label |
|---|---|
| 3 | Input: prompt to build the material catalog |
| 5 | Input: prompt to build the layup catalog |
| 6 | Input: prompt to build the cross-section catalog |
| 7 | Input: prompt to model a tape spring under a 1 N tip load |
| 9 | Input: prompt to verify the same tape spring in Abaqus |
| 11 | Input: prompt to create the toy catalogs (unchanged) |
| 12 | Input: prompt to run the tools on the known optimum (unchanged) |
| 13 | Input: prompt for the lightest boom |
| 15 | Input: prompt to reproduce a paper |

On slide 13 I wanted "to find the lightest boom", but that wrapped onto the image below it, so it's shortened to "for the lightest boom".

User · 09:34

Similarly, for outputs, do "Output: Brief description" for each

**Bash** Find all output labels

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n '"Output\|Output:\|out_label\|"Sample rows'
build_demo_deck.py
```

```
8:Output: "Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx".
371:    def catalog_slide(title, crop, desc, out_label, cols, rows, notes, extra=None, ref=None, row_h=0.45,
390:            label(s, X, y, W, out_label)
415:        "Sample rows of the material catalog",
423:        "Sample rows are copied from .claude/memory.material_property_catalog/UD_lamina_catalog_readable.txt. "
```

```
433:        ref_y=6.6, io_label="Output")
440:        "Output: sample rows of the layup catalog",
449:        "Sample rows from .claude/memory.layup_catalog/layup_catalog.txt (199 rows: 15 with 3 plies, 15 with 4, "
469:        "Output: sample rows of the cross-section catalog",
477:        "Sample rows from .claude/memory.boom_geometry_catalog/boom_geometry_catalog.txt. Size = width x height of "
[... shortened ...]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-08.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-14.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-15.png

```
[image]
```

**Bash** Locate slide 14 (pruning grid) layout code

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n "# 1[0-9] ---\|# 9 ---" build_demo_deck.py && sed -n
'/# 13 ---/,/# 14 ---/p' build_demo_deck.py | grep -n "add_\|label\|y0\|Y0\|= [0-9]" | head -50
```

```
585:    # 9 ------------------------------------------------------------------ Abaqus loop: results
622:    # 10 ----------------------------------------------------------------- baseline: sandbox
660:    # 11 ----------------------------------------------------------------- baseline: tools only
704:    # 12 ----------------------------------------------------------------- baseline: orchestrator
739:    # 13 ----------------------------------------------------------------- baseline: pruning
822:    # 14 ----------------------------------------------------------------- failure case
7:                  "theory (k = 3 E_x I / L^3, rejected below 0.85 x 0.05 N/mm). 17 fail coiling, 12 fail stiffness "
8:                  "(some both): every L003 row (strengths x0.001), every T002 row (R = 2 mm) and [0]10 on R = 20 mm "
9:                  "fail coiling; [90/90/90] on R = 20 mm and every R = 2 mm row fail stiffness. Mass was not used to "
12:                  "already the optimum; the other 7 feasible designs are all heavier (R = 200 mm or 1000x density).\n"
[... shortened ...]
```

**Bash** Show slide 14 grid header code

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -n 754,766p build_demo_deck.py
```

```
gx, gy, tw, th, gap = 1.35, 1.62, 0.66, 0.72, 0.06
    layups = ["[0/0/0]", "[0]_{10}", "[90/90/90]"]
    geos = ["20", "2", "200"]
    for li, ly in enumerate(layups):
        x0 = gx + li * (3 * (tw + gap) + 0.12)
        add_text(s, x0, 0.8, 3 * (tw + gap), 0.4, [[(ly, {"bold": True})]], size=18, align=PP_ALIGN.CENTER,
                 space_after=0)
        for gi, g in enumerate(geos):
            add_text(s, x0 + gi * (tw + gap) - 0.05, 1.17, tw + 0.1, 0.4, g, size=18, align=PP_ALIGN.CENTER,
                     space_after=0)
[... shortened ...]
```

**Bash** Add output labels, shift layouts, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && cat >
/c/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/e10.py
<<'EOF'
f = "build_demo_deck.py"
s = open(f, encoding="utf8").read()
def rep(a, b):
    global s
    assert s.count(a) == 1, a[:70]
    s = s.replace(a, b)
# catalog slides: no table label when out_label is None (slide 4 has it in the top label)
rep('''        if rows:
            label(s, X, y, W, out_label)
            add_table(s, X, y + 0.43,''', '''        if rows:
            if out_label:
                label(s, X, y, W, out_label)
                y += 0.43
            add_table(s, X, y,''')
rep('''        "Sample rows of the material catalog",''', '''        None,''')
rep('''        ref_y=6.6, io_label="Output")''', '''        ref_y=6.6, io_label="Output: sample rows of the material
catalog")''')
```

```
# slide 10: label + shift left column down
[... shortened ...]
```

```
ok
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 1: text overflow 0.05 in in 'TextBox 2': 'AI Agents for the Optimization of\nThin-Shell Compo'
slide 4: table 'Table 11' grows 0.42 in past set height, bottom at 3.97 in
slide 6: table 'Table 14' grows 0.23 in past set height, bottom at 6.51 in
slide 10: table 'Table 7' grows 0.23 in past set height, bottom at 4.06 in
slide 12: text overflow 0.09 in in 'TextBox 13': 'Strength ratio R = 1/√F: factor by which the load '
slide 15: table 'Table 10' grows 1.06 in past set height, bottom at 4.76 in
All validations PASSED!
```

Claude · 09:35

Every slide with an output now has an "Output: …" label and the deck rebuilds cleanly. Next I'm checking the renders of slides 4, 10, 14 and 15, where labels were added or changed.

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-14.png

```
[image]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-04.png

```
[image]
```

**wrap_up** wrap-up

```
Every slide with an output now has an "Output: …" label. Slides 4, 10, 14 and 15 got new or reworded labels, and the slide
10 and 14 content moved down to make room. The validator passes; not checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 09:36

Every slide that shows an output now has an "Output: …" label. Most already did, so only four changed:

| Slide | Was | Now |
|---|---|---|
| 4 | "Output" at the top, plus "Sample rows of the material catalog" above the table | Output: sample rows of the material catalog (one label; the table moved up) |
| 10 | no label | Output: OpenSees vs. Abaqus comparison (above the table) |
| 14 | no label | Output: pre-screening of the 27 possible designs (above the grid) |
| 15 | "Output" | Output: expected vs. returned |

On slides 10 and 14 the left-hand content moved down a little to make room. The renders show nothing overlapping.

The other output labels stay as they were: - Slide 5: Output: sample rows of the layup catalog - Slide 6: Output: sample rows of the cross-section catalog - Slide 7: Output: tip response, mesh convergence, beam theory - Slide 8: Output: objective/constraint check - Slide 9: Output: OpenSeesPy and Abaqus agree within 0.7% (next slide) - Slide 11: Output: 3 × 3 × 3 = 27 possible designs - Slide 12: Output: all three load cases pass - Slide 13: Output: optimal design vs. baseline vs. Abaqus

I rebuilt the deck and it passes the file validator; I haven't opened it in PowerPoint.

User · 09:37

Slide 10: How does sigma_1 for top ply differ from sigma_xx?

Claude · 09:37

For this model they are the same stress, just written in two coordinate systems.

- **$\sigma_{xx}$** is the normal stress along the boom axis x, in the global (boom) coordinates.
- **$\sigma_1$** is the normal stress along the fibers, in the ply's own material axes. Hashin failure is checked in these axes.

The two are related by the ply angle θ:

$\sigma_1 = \sigma_{xx} \cos^2\theta + \sigma_{yy} \sin^2\theta + 2\tau_{xy} \sin\theta \cos\theta$

This demo uses a [0/0/0] layup, so θ = 0, the fibers run along x, and $\sigma_1 = \sigma_{xx}$ at every point. The table is consistent with that: the $\sigma_1$ row (46.14 / 46.33 MPa) equals the top of the $\sigma_{xx}$ range (46.1 / 46.3 MPa). The $\sigma_{xx}$ picture titles also give the same maxima, 46.14 and 46.33.

The rows differ only in what they report: - **$\sigma_1$ row:** only the peak tensile fiber stress, which drives the fiber-tension failure mode. - **$\sigma_{xx}$ row:** the full range, including the compressive minimum (about −29.6 MPa), and it matches the colored pictures.

For a layup with off-axis plies, such as [45/−45/0/−45/45], the two would differ.

If the two rows look redundant on the slide, I can merge them into one row, for example "$\sigma_{xx} = \sigma_1$ (0° plies), top ply: −29.6 to 46.1". Or I can add a short note under the table saying they coincide for 0° plies.

User · 09:39

If sigma_xx and sigma_1 coincide here, call them both sigma_1

**Bash** Find sigma labels in build script

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n 'σ_{xx}\|σ_{1}' build_demo_deck.py
```

```
535:    add_text(s, 0.3, 0.75, 6.2, 0.4, [[("Axial stress σ_{xx} on the top ply from bending", {"bold": True})]],
601:                ["σ_{1}, top ply (MPa)", "46.14", "46.33", "0.4%"],
603:                ["σ_{xx} range, top ply (MPa)", "−29.6 to 46.1", "−29.8 to 46.3", "< 1%"]],
619:    for key, nm, t in (("img12", "compare_uz", "u_{z}"), ("img11", "compare_sxx", "σ_{xx}")):
732:                ["Peak σ_{1} at root (MPa)", "50.0", "47.7", "42.6 (top ply)"],
```

**Bash** Rename sigma_xx to sigma_1 on slide 10, rebuild

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i '601s/"σ_{1}, top ply (MPa)"/"Max σ_{1}, top ply (MPa)"/; 603s/"σ_{xx} range, top ply (MPa)"/"σ_{1} range, top ply (MPa)"/; 619s/"σ_{xx}")/"σ_{1}")/' build_demo_deck.py && sed -n '601p;603p;619p' build_demo_deck.py && SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 10"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
Command was moved to the background (ID: btrj14lm7) so that a message that arrived while it was running can reach you; it was not interrupted. Output is being written to: C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\07583a8d-5e34-4ade-abc5-3c5983a9a11f\tasks\btrj14lm7.output. You will be notified when it completes. It is running now: do not start a second copy while it runs. If it is no longer needed, stop it with TaskStop. To check interim output, use Read on that file path.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3; directory changes made by the backgrounded command do not apply to subsequent commands.
```

Claude · 09:40

Okay, I'll remove the range row. The rebuild with the $\sigma_1$ renaming is still running in the background, so I'll edit once it finishes.

**Bash** Wait for background rebuild and show output

```
sleep 20; cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/07583a8d-5e34-4ade-abc5-3c5983a9a11f/tasks/btrj14lm7.output"
```

```
["Max σ_{1}, top ply (MPa)", "46.14", "46.33", "0.4%"],
              ["σ_{1} range, top ply (MPa)", "−29.6 to 46.1", "−29.8 to 46.3", "< 1%"]],
    for key, nm, t in (("img12", "compare_uz", "u_{z}"), ("img11", "compare_sxx", "σ_{1}")):
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 10: table 'Table 7' grows 0.23 in past set height, bottom at 4.06 in
All validations PASSED!

[exited with code 0]
```

**Bash** Delete range row, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i '603{/σ_{1} range, top ply/d}' build_demo_deck.py && sed
-i '602s/\],$/]],/' build_demo_deck.py && sed -n 598,604p build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 10"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-
mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts";
"$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting
Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
[["Quantity", "OpenSees", "Abaqus", "Diff."],
              ["Tip u_{z} (mm)", "−8.523", "−8.583", "0.70%"],
              ["Tip rotation (rad)", "0.01266", "0.01275", "0.70%"],
              ["Max σ_{1}, top ply (MPa)", "46.14", "46.33", "0.4%"],
              ["τ_{12}, top ply (MPa)", "0.355", "0.354", "0.1%"]],
             size=18, row_h=0.45, first_col_bold=True)
    add_box(s, 0.3, 4.4, 6.1, 1.5,
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**Read** C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/r/s-10.png

```
[image]
```

**Claude · 09:40**

With the range row gone there's an empty gap below the table, so I'll move the "Agree within 0.7%" box and the Working pill up to close it.

**Bash** Close the gap on slide 10, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/    add_box(s, 0.3, 4.4, 6.1, 1.5,/    add_box(s,
0.3, 4.0, 6.1, 1.5,/; s/    status_pill(s, 0.3, 6.12)/    status_pill(s, 0.3, 5.75)/; s/    add_text(s, 2.15, 6.09,/
add_text(s, 2.15, 5.72,/' build_demo_deck.py && grep -c "0.3, 4.0, 6.1, 1.5\|status_pill(s, 0.3, 5.75)\|2.15, 5.72"
build_demo_deck.py && SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-
3c5983a9a11f/scratchpad"; V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V"
"$SP/render.py" "Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 10";
SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-
43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3
Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite
Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
3
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
Slide 10 now calls the stress σ₁ everywhere: the range row is gone, the row label reads "Max σ₁, top ply", and the lower
pictures are labeled σ₁. The validator passes; not checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

**Claude · 09:41**

I removed the range row from the slide 10 table, and the stress is now called $\sigma_1$ throughout the slide: - **Table:** the stress row reads "Max $\sigma_1$, top ply (MPa)". I added "Max" because 46.14 / 46.33 MPa is the peak value, not a

point value like the tip numbers. - **Pictures:** the label beside the lower pair now reads $\sigma_1$. - **Layout:** I moved the "Agree within 0.7%" box and the Working pill up to fill the space the row left.

I rebuilt [the deck] and it passes the file validator.

Two places still say $\sigma_{xx}$: - **The pictures themselves:** the small captions and colorbar titles inside the screenshots read "sigma_xx", and they can't be edited. - **Slide 8:** its plot title says "Axial stress $\sigma_{xx}$ on the top ply from bending". It's the same [0/0/0] tape spring, so $\sigma_1$ would be correct there too. Tell me if you want that changed for consistency.

User · 09:45

Slide 8: Replace "objective/constraint check" with "mass and material failure index calculator

**Bash** Replace slide 8 output label, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/"Output: objective\/constraint check"/"Output: mass and material failure index calculator"/' build_demo_deck.py && grep -n "mass and material failure index calculator" build_demo_deck.py && SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 8"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
541:    label(s, 0.3, 4.85, 6.0, "Output: mass and material failure index calculator")
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
Slide 8's output label now reads "Output: mass and material failure index calculator". The deck is rebuilt and the validator passes; not checked in PowerPoint.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

Claude · 09:45

Slide 8's label now reads "Output: mass and material failure index calculator". It fits on one line in the space it had. I rebuilt [the deck] and it passes the file validator.

User · 09:45

Get rid of speaker notes -- this will be saved as a PDF eventually and not presented, so they are unnecessary

**Bash** Find where speaker notes are written

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n "if notes:" -A1 build_demo_deck.py
```

```
328:        if notes:
329-            s.notes_slide.notes_text_frame.text = notes
```

**Bash** Stop writing notes, rebuild, verify none remain

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i '328,329d' build_demo_deck.py && sed -i '327a\        # notes (presenter, sources) are kept here for reference only: the deck is shared as a PDF, so no\n        # speaker notes are written' build_demo_deck.py && sed -n 322,332p build_demo_deck.py &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && "$V" -c "
from pptx import Presentation
p=Presentation('Lab Report 3 Demonstration of Agent System.pptx')
print('slides with notes:', sum(s.has_notes_slide and s.notes_slide.notes_text_frame.text.strip()!='' for s in p.slides))
" && SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3
```

```
Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
blank = [l for l in prs.slide_layouts if l.name == "Blank"][0]

    def new_slide(title=None, notes=None):
        s = prs.slides.add_slide(blank)
        if title:
            header(s, title, len(prs.slides), logo)
        # notes (presenter, sources) are kept here for reference only: the deck is shared as a PDF, so no
        # speaker notes are written
        return s

[... shortened ...]
```

**wrap_up** wrap-up

```
The deck no longer has speaker notes on any of its 15 slides. The note text stays in the build script for reference but is no longer written to the deck; the rebuilt deck passes the validator.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

**Claude · 09:46**

I removed the speaker notes, so none of the 15 slides has notes now. I rebuilt the deck and it passes the file validator.

The note text is still in build_demo_deck.py, next to each slide's code: suggested presenters, the catalog file paths, and details such as the Abaqus mesh and buckling numbers. The script just no longer puts it in the deck. That's handy if you want any of those details in the written report.

**User · 09:48**

Slide 13: change the input to "prompt for boom optimization"

**Bash** Change slide 13 input label, rebuild, validate

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i 's/"Input: prompt for the lightest boom"/"Input: prompt for boom optimization"/' build_demo_deck.py && grep -n "prompt for boom optimization" build_demo_deck.py && SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad"; V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3 Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 13"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts"; "$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
718:    label(s, 0.3, 0.78, 4.2, "Input: prompt for boom optimization")
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
All validations PASSED!
[This command modified 1 file you've previously read: build_demo_deck.py. Call Read before editing.]
```

**wrap_up** wrap-up

```
Slide 13's input label now reads "Input: prompt for boom optimization" and fits on one line. The deck is rebuilt and the validator passes.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the user: write it now, as the last thing you say this turn.
```

**Claude · 09:48**

Slide 13's label now reads "Input: prompt for boom optimization", and it fits on one line. I rebuilt the deck and it passes the file validator.

**User · 09:53**

Final slide: Make sure reference numbering is consistent with the rest of the slide deck

**Bash** List all reference numbers in deck

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && grep -n '\[[0-9]\]' build_demo_deck.py | cut -c1-160
```

```
78:    s = im.size[0] / 722.0  # coordinates below were read on a 722 px wide copy
88:            h = [h[0] - b[0], h[1] - b[1], h[2] - b[0], h[3] - b[1]]
99:    im.crop((0, 0, im.size[0], min(im.size[1], bbox[3] + int(10 * s)))).save(full, "PNG")
156:        p = tf.paragraphs[0] if i == 0 else tf.add_paragraph()
181:        sh.adjustments[0] = adj
204:            p = tf.paragraphs[0] if i == 0 else tf.add_paragraph()
215:    ar = im.size[1] / im.size[0]
232:    nr, nc = len(rows), len(rows[0])
259:            p = tf.paragraphs[0]
322:    blank = [l for l in prs.slide_layouts if l.name == "Blank"][0]
[... shortened ...]
```

**Bash** Renumber final-slide reference to [7], rebuild

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Lab Report 3" && sed -i '827s/Palmeri and Laurenzi \[1\]"/Palmeri and Laurenzi
[7]"/; 855s/"\[1\] Palmeri/"[7] Palmeri/' build_demo_deck.py && sed -n '827p;855p' build_demo_deck.py && grep -n "\[1\]"
build_demo_deck.py | sed -n '1,20p' | cut -c1-120 &&
SP="C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad";
V="$SP/venv/Scripts/python.exe"; "$V" build_demo_deck.py && PYTHONIOENCODING=utf8 "$V" "$SP/render.py" "Lab Report 3
Demonstration of Agent System.pptx" "$SP/r/s" | grep "slide 15:"; SK="C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-
mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts";
"$V" "$SK/office/validate.py" "Lab Report 3 Demonstration of Agent System.pptx" --original "../11-11-26 Group Meeting
Presentation/AI Agents for Thin-Shell Composite Booms (11-11-26).pptx" 2>&1 | tail -1
```

```
s = new_slide("Failure Case: Omega Boom of Palmeri and Laurenzi [7]",
    refs(s, ["[7] Palmeri, F., & Laurenzi, S. (2024). Neural network-based surrogate modeling for buckling "
88:            h = [h[0] - b[0], h[1] - b[1], h[2] - b[0], h[3] - b[1]]
99:    im.crop((0, 0, im.size[0], min(im.size[1], bbox[3] + int(10 * s)))).save(full, "PNG")
215:    ar = im.size[1] / im.size[0]
420:         ["L001", "IM7/8552 epoxy [1]", "158.5/8.95/4.67", "2501/64.0/91.1", "1.58"],
428:        ref=["[1] NCAMP (2011). Hexcel 8552 IM7 unidirectional prepreg qualification material property data report,
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Lab Report 3\Lab Report 3 Demonstration of Agent System.pptx 15 slides
slide 15: table 'Table 10' grows 1.06 in past set height, bottom at 4.76 in
All validations PASSED!
[... shortened ...]
```

**wrap_up** wrap-up

```
The final slide's reference is now [7], in its title and its reference line, continuing the [1]-[6] numbering of slides 4
and 6. The deck is rebuilt and the validator passes.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 09:53

The Palmeri and Laurenzi paper on the final slide is now [7], both in the slide title and in its reference line. The numbering now runs straight through the deck:

| Numbers | Slide | Sources |
|---|---|---|
| [1]–[4] | 4 (material catalog) | NCAMP, Soden et al., Liu et al., Toray |
| [5]–[6] | 6 (cross-section catalog) | Benazzo & Pellegrino, Fernandez & Volle |
| [7] | 15 | Palmeri & Laurenzi |

I rebuilt the deck and it passes the file validator.

User · 11:47

Create a PDF of this conversation in the Claude Conversations folder. This is today's conversation 1

**[10:48] Context compacted.** The session continued from this automatic summary:

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.

Summary: 1. Primary Request and Intent: - Overall: build and refine the Lab Report 3 demo deck, `Lab Report 3/Lab Report 3 Demonstration of Agent System.pptx`, in the style of the 11-11-26 group meeting deck. Earlier history (title slide, system diagram, demo slides, revision rounds, the diagram update for Claude running Abaqus on the server, removing ", by Claude") was completed before the earlier compaction. - Requests since then, all done: - Put Federico's plain-text librarian prompts (material, layup and geometry catalogs) into prompt bubbles that look like the user's Claude-prompt screenshots. Mid-task the user added: "Don't split the list of columns into 2 columns -- make it look like a continuous Claude prompt". - Remove "Federico's prompt to the librarian agent". - Put boxes around the bubbles, like the screenshots on the later slides. - Remove "(Input)" and "(Output)" from the slide 3 and 4 titles. - Five-part revision: - Move the slide 3 bubble down so it doesn't overlap the "Working" pill. - Slide 4: remove "* = estimate" from the references. - Slide 9: crop the diagram so the bottoms of the boxes above the verification loop aren't visible. - Slide 10: label the 4 figures OpenSees/Abaqus and u_z/σ_xx, with subscripts, at 18 pt, keeping the small in-figure descriptions. - All prompt inputs read "Input: prompt to ...". - Outputs read "Output: brief description" on every slide. - Answered a question on how σ_1 and σ_xx differ. They are the same for the [0/0/0] layup; the user then said to call both σ_1, and mid-task: "Actually just eliminate the range row". - Slide 8: label reads "Output: mass and material failure index calculator". - Remove all speaker notes; the deck will be saved as a PDF and not presented. - Slide 13 input: "prompt for boom optimization". - Final slide: make the reference numbering consistent with the rest of the deck. - Constraints from CLAUDE.md and the organization: - Run Abaqus only via run_abaqus.py, with at most 8 CPUs. - Never write anything outside the ME295 folder on the shared server. - Protect sensitive data; never share the user's email with external services. - Commit only if the user asks. Nothing has been committed.

2. Key Technical Concepts: - The deck is built with python-pptx on top of the 11-11 deck's masters and theme (all of its slides are dropped). Style: grey header with an orange rule, Calibri, body text at least 18 pt, Caltech logo at x ≥ 11.64 in, y ≥ 6.82 in. - `_{...}` markup becomes subscript runs. - Prompt bubbles are PIL-rendered PNGs: Segoe UI, background (240,240,240), rounded corners, 300 dpi, white margin. On the slide they get a grey BFBFBF 0.75 pt picture border, like the screenshots. - QA:
   - A custom PIL renderer stands in for LibreOffice or PowerPoint, neither of which is installed. It has known quirks, e.g. subscripts are drawn raised.
   - The pptx skill's validate.py runs with `--original` set to the 11-11 deck.
   - Paths:
   - Venv python: `C:/Users/tvenkata/AppData/Local/Temp/16/claude/91fa69c9debc8176/07583a8d-5e34-4ade-abc5-3c5983a9a11f/scratchpad/venv/Scripts/python.exe`
   - Renderer: `$SP/render.py deck $SP/r/s`, run with PYTHONIOENCODING=utf8 because of the √ character.
   - Validator: `C:/Users/tvenkata/AppData/Roaming/Claude/local-agent-mode-sessions/skills-plugin/94279154-d6dd-451c-942c-43fed0664975/d1887321-156e-46f7-896e-5ac22acf505e/skills/pptx/scripts/office/validate.py`
   - Edits are made with sed, or with small Python `rep()` scripts in the scratchpad (e5–e10.py) that assert each target appears exactly once.
3. Files and Code Sections: - `Lab Report 3/prompt_bubbles.py` (new):
   - Holds MATERIAL, LAYUP and GEOMETRY item lists, verbatim. Item types: `("p", text)`, `("b", text)`, `("n", "1.", text)`, `("gap",)`, `("cols", lines, ncol)`. The material column list now uses ncol=1, and the extra "gap" before the last paragraph was removed.
   - `bubble(items, width_in, font_pt, dpi=300, margin_in=0.08)` returns PNG bytes: the grey rounded bubble pasted onto a white RGB image with a margin.
   - `Lab Report 3/build_demo_deck.py`:
   - Imports `prompt_bubbles as pb`.
   - `catalog_slide(title, crop, desc, out_label, cols, rows, notes, extra=None, ref=None, row_h=0.45, ref_y=6.93, prompt=None, io_label="Input / Output Example", prompt_pt=8.5, input_label="Input")`:
   - The top label is drawn with width 6.0, at 20 pt if there is a table and 18 pt otherwise.
   - On a prompt-only slide the bubble starts at y=1.33. Otherwise the input label sits at y=1.3 and the bubble below it at +0.42. The bubble is placed with `add_pic(..., border=True, name="prompt_librarian")`.
   - The out_label is drawn only if it is truthy.
   - There are 15 slides: 3. Material input: io_label "Input: prompt to build the material catalog", prompt_pt=8.25; bubble spans 1.33–6.74. 4. Material output: io_label "Output: sample rows of the material catalog", out_label None; ref [4] no longer ends with "* = estimate". 5. Layup: input_label "Input: prompt to build the layup catalog". 6. Geometry: input_label "Input: prompt to build the cross-section catalog"; column widths [0.8, 1.75, 2.75, 1.38, 1.2], row_h 0.42. 7. Label "Input: prompt to model a tape spring under a 1 N tip load". 8. Label "Output: mass and material failure index calculator". 9. Label "Input: prompt to verify the same tape spring in Abaqus"; the br crop is now `crop((350, 302, 722, 528), white=(0, 760, 75, 900))`. 10. Label "Output: OpenSees vs. Abaqus comparison" at y 0.72; table at y 1.14 with rows Tip u_{z}, Tip rotation, "Max σ_{1}, top ply (MPa)" and τ_{12} (the range row was deleted); box at (0.3, 4.0, 6.1, 1.5); pill at 5.75; italic text at y 5.72. Pictures at PX=7.1, PW=5.95, with "OpenSees"/"Abaqus" headers at y 0.72 and row labels "u_{z}" and "σ_{1}" at x 6.45. 13. Label "Input: prompt for boom optimization". 14. Label "Output: pre-screening of the 27 possible designs"; grid shifted down by dy=0.4. 15. Label "Output:

expected vs. returned" (width 6.6, 18 pt); title "Failure Case: Omega Boom of Palmeri and Laurenzi [7]"; reference line "[7] Palmeri, F., & Laurenzi, S. (2024)...".
    - Slide 2 checklist ranges: 3-6, 7-8, 9-10, 11-14, 15. The sandbox slide says "L001 (slide 4)"; the failure slide says "slides 11-14".
    - `new_slide` no longer writes notes. A comment explains that the notes are kept in the code for reference only.
    - Output: the deck, now 15 slides.
4. Errors and fixes: - The material prompt was too tall for the right column, so the material catalog was split across slides 3 (input) and 4 (output), and slide numbers were updated everywhere. - An e7.py replacement failed its count assertion because the pattern was ambiguous; fixed with more specific text. - In the renderer, the slide 3 label wrapped (fixed by widening it and using 18 pt), the slide 6 table hit the logo (fixed with column widths and row height), the slide 3 bubble touched the logo (fixed with a smaller margin and font), and the slide 13 label wrapped (fixed by shortening it). - The renderer crashed printing √ in the console; fixed with PYTHONIOENCODING=utf8. - One command was moved to the background when a user message arrived; waited for it to finish before editing. - Known renderer warnings that are not regressions: slide 1 title (0.05 in), slide 4 header wrap, slide 12 text (0.09 in), slide 15 table growth.
5. Problem Solving: - All requested edits are applied, and the validator passes after each rebuild. - Nothing has been checked in real PowerPoint. - Two places still say σ_xx, as told to the user: the captions and colorbars inside the slide 10 screenshots, and the slide 8 plot title. The user did not ask to change slide 8.
6. All user messages: - "These are Federico's prompts to be put on the material/layup/geometry slides. He gave them to me as plain text instead of screenshots, so please format them into the text bubbles so they look like my screenshots." Followed by the full verbatim texts of the material, layup and geometry prompts. - "Don't split the list of columns into 2 columns -- make it look like a continuous Claude prompt" - "What are you still doing?" - "Get rid of where it says "Federico's prompt to the librarian agent"" - "And put boxes around the bubbles like in the later slides" - "And get rid of (Input) and (Output) in the slide 3/4 titles" - "Move the box with the screenshot down on slide 3 so it doesn't overlap the "Working" bubble / Slide 4: Remove the * = estimate from the bottom of the slide because it's already in the "what this section does" / Slide 9: crop the screenshot of the system diagram so the bottom of the boxes above the verification loop boxes are not visible / Slide 10: For the 4 figures, label OpenSees and Abaqus, and u_z and sigma_xx (with subscripts), in 18-pt font so it's clear what each figure represents (but keep the little descriptions in the top left of each figure since they have more detail) / For all slides where the input is a prompt, say "Input: prompt to ..." where ... describes what the prompt does in a few words" - "Similarly, for outputs, do "Output: Brief description" for each" - "Slide 10: How does sigma_1 for top ply differ from sigma_xx?" - "If sigma_xx and sigma_1 coincide here, call them both sigma_1" - "Actually just eliminate the range row" - "Slide 8: Replace "objective/constraint check" with "mass and material failure index calculator" - "Get rid of speaker notes -- this will be saved as a PDF eventually and not presented, so they are unnecessary" - "Slide 13: change the input to "prompt for boom optimization"" - "Final slide: Make sure reference numbering is consistent with the rest of the slide deck"
7. Pending Tasks: - None explicitly pending. - Possible follow-up the user hasn't answered: change slide 8's "σ_xx" plot title to σ_1. Do not do this unless asked. - Commit only if asked.
8. Current Work: - The last request was "Final slide: Make sure reference numbering is consistent with the rest of the slide deck". - Changed line 827 to `s = new_slide("Failure Case: Omega Boom of Palmeri and Laurenzi [7]",` and line 855 to start the reference with `"[7] Palmeri, F., & Laurenzi, S. (2024)...`. This continues the [1]–[4] references on slide 4 and [5]–[6] on slide 6. - Rebuilt; the validator passed. - wrap_up was called with the recap. The closing text message to the user had not yet been written.
9. Optional Next Step: - Send the brief closing message: the final slide's reference is now [7] in both its title and its reference line, continuing [1]–[6] from slides 4 and 6; the deck is rebuilt and the validator passes. - No further work unless the user asks.

If you need specific details from before compaction (like exact code snippets, error messages, or content you generated), read the full transcript at: C:\Users\tvenkata.claude\projects\F--Tara-BenazzoVenkatadri-ThinShellSims\07583a8d-5e34-4ade-abc5-3c5983a9a11f.jsonl Continue the conversation from where it left off without asking the user any further questions. Resume directly — do not acknowledge the summary, do not recap what was happening, do not preface with "I'll continue" or similar. Pick up the last task as if the break never happened.

The messages after this summary are the most recent messages from before compaction, kept verbatim. The summary was written without seeing them, so something it says has not happened yet may already have happened in them.

**update_plan** task list

```
[in_progress] Find Claude Conversations folder and conventions
[pending] Convert transcript to PDF
[pending] Check the PDF
```

```
Plan shown (3 steps).
```

**Bash** Locate Claude Conversations folder and list contents

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; find . -maxdepth 3 -iname "*Claude Conv*" -not -path "./.git/*"; ls "Claude Conversations" 2>/dev/null | tail -30
```

```
./Claude Conversations
Federico 11-5-26 Conversation 1 (Orchestrator v3.0 - protocol, design screen, study manager, hook).pdf
Tara 10-23-26 Conversation 1 (Python setup and Federico's 10-22 test cases).pdf
Tara 10-23-26 Conversation 2 (Building material property agents).pdf
Tara 10-23-26 Conversation 3 (Testing material property agents).pdf
Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf
Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf
Tara 11-11-26 Conversation 1 (Baseline with tools only, no orchestrator).pdf
Tara 11-11-26 Conversation 2 (Optimization to replicate baseline).pdf
Tara 11-11-26 Conversation 3 (System component demos for Lab Report 3).pdf
[... shortened ...]
```

**Bash** Look for existing transcript-to-PDF scripts and the transcript file

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims"; git log --oneline -3 -- "Claude Conversations"; grep -rl --include=*.py -i
"jsonl" . 2>/dev/null | grep -v .git | head; ls ~/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/*.jsonl | head
-30; ls -la ~/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/07583a8d-5e34-4ade-abc5-3c5983a9a11f.jsonl
```

```
Exit code 2
164db58 Lab 3 narrative report and demo slides draft
f5e9bfb 11-11-26 (Tara): tape-spring baseline - tools-only run, sandboxed orchestrator toy study, report and comparison
e82bd92 11-10-26 (Tara): CTM omega boom setup - P&L ABD fit, exact laminates, mesh and Abaqus buckling checks
./Code for Project/11-11-26 (Tara)/conversation_to_pdf.py
ls: cannot access '/n//.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/*.jsonl': No such file or directory
ls: cannot access '/n//.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/07583a8d-5e34-4ade-abc5-3c5983a9a11f.jsonl':
No such file or directory
```

**Bash** Read the existing conversation-to-PDF script header

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; sed -n 1,60p conversation_to_pdf.py; wc -l
conversation_to_pdf.py; ls -la "/c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/" | tail -8
```

```
(.*?) """Turn a Claude Code session transcript (.jsonl) into an HTML page and print it to PDF
with headless Chrome (same route as the earlier Claude Conversations PDFs).

Run with the system Python (needs `markdown`, installed in ./_pylibs):
    python conversation_to_pdf.py <transcript.jsonl> "<output.pdf>" "<title>"
Shows user messages and assistant text in full, tool calls and their results shortened;
thinking blocks and system reminders are left out."""
import html
import json
import re
[... shortened ...]
```

**Bash** Read rest of script and check transcript and python

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; sed -n 60,161p conversation_to_pdf.py; ls -la
/c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/07583a8d*; which python; python -c "import
sys;print(sys.version)"
```

```
body = '\n'.join(f"[{i['status']}] {i['content']}" for i in items) if items else json.dumps(inp)
    elif name == 'mcp__ccd_turn__wrap_up':
        head, body = 'wrap-up', inp.get('recap', '')
    else:
        head, body = '', short(json.dumps(inp, ensure_ascii=False, indent=1), 10, 1000)
    label = name.replace('mcp__ccd_turn__', '').replace('mcp__', '')
    out = f'<div class="tool"><span class="tn">{html.escape(label)}</span> {html.escape(head)}'
    if body:
        out += f'<pre>{html.escape(body)}</pre>'
    return out + '</div>'
[... shortened ...]
```

**Bash** Check session title and time span

```
cd /c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/; cat 07583a8d-5e34-4ade-abc5-
3c5983a9a11f/custom-title.json; echo; grep -o '"timestamp":"[^"]*"' 07583a8d-5e34-4ade-abc5-3c5983a9a11f.jsonl | sed -n
'1p;$p'
```

```
{"customTitle":"Lab Report 3 slide deck"}
"timestamp":"2026-11-12T02:39:23.183Z"
"timestamp":"2026-11-12T19:48:04.078Z"
Shell cwd was reset to F:\Tara\BenazzoVenkatadri_ThinShellSims
```

Claude · 11:48

Generating the PDF now with the same script that made the earlier conversation PDFs.

**Bash** Convert this session transcript to PDF in Claude Conversations

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)" && python - <<'EOF'
p='conversation_to_pdf.py'; s=open(p,encoding='utf8').read()
a="    span = f'{first_ts:%H:%M} - {last_ts:%H:%M} PST' if first_ts else ''"
b="""    span = f'{first_ts:%H:%M} - {last_ts:%H:%M} PST' if first_ts else ''
    if first_ts and last_ts and last_ts.date() != first_ts.date():   # session ran past midnight
        span = f'{first_ts:%H:%M} - {last_ts:%m-%d-%Y %H:%M} PST'"""
assert s.count(a)==1; open(p,'w',encoding='utf8').write(s.replace(a,b)); print('ok')
EOF
python conversation_to_pdf.py "/c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/07583a8d-5e34-
4ade-abc5-3c5983a9a11f.jsonl" "../../Claude Conversations/Tara 11-12-26 Conversation 1 (Lab Report 3 demo slide deck).pdf"
"Tara 11-12-26 Conversation 1: Lab Report 3 demo slide deck"
```